Neural Computation -- 2013

102 article(s), all issues

1. Encoding Binary Neural Codes in Networks of Threshold-Linear Neurons

Amos Maritan

(no DOI)

No abstract available.

---

2. Neural Representation of Spatial Topology in the Rodent Hippocampus

Zhe Chen, Stephen N. Gomperts, J. Yamamoto, Matthew A. Wilson

https://doi.org/10.1162/neco_a_00538 | pp. 1-39

Pyramidal cells in the rodent hippocampus often exhibit clear spatial tuning in navigation. Although it has been long suggested that pyramidal cell activity may underlie a topological code rather than a topographic code, it remains unclear whether an abstract spatial topology can be encoded in the ensemble spiking activity of hippocampal place cells. Using a statistical approach developed previously, we investigate this question and related issues in greater detail. We recorded ensembles of hippocampal neurons as rodents freely foraged in one- and two-dimensional spatial environments and used a "decode-to-uncover" strategy to examine the temporally structured patterns embedded in the ensemble spiking activity in the absence of observed spatial correlates during periods of rodent navigation or awake immobility. Specifically, the spatial environment was represented by a finite discrete state space. Trajectories across spatial locations ("states") were associated with consistent hippocampal ensemble spiking patterns, which were characterized by a state transition matrix. From this state transition matrix, we inferred a topology graph that defined the connectivity in the state space. In both one- and two-dimensional environments, the extracted behavior patterns from the rodent hippocampal population codes were compared against randomly shuffled spike data. In contrast to a topographic code, our results support the efficiency of topological coding in the presence of sparse sample size and fuzzy space mapping. This computational approach allows us to quantify the variability of ensemble spiking activity, examine hippocampal population codes during off-line states, and quantify the topological complexity of the environment.

---

3. Single-Snippet Analysis for Detection of Postspike Effects

Sagi Perel, Andrew B. Schwartz, Valérie Ventura

https://doi.org/10.1162/neco_a_00531 | pp. 40-56

Corticomotoneuronal cells (CMN), located predominantly in the primary motor cortex, project directly to alpha motoneuronal pools in the spinal cord. The effects of CMN spikes on motoneuronal excitability are traditionally characterized by visualizing postspike effects (PSEs) in spike-triggered averages (SpTA; Fetz, Cheney, & German, 1976; Fetz & Cheney, 1980; McKiernan, Marcario, Karrer, & Cheney, 1998) of electromyography (EMG) data. Poliakov and Schieber (1998) suggested a formal test, the multiple-fragment analysis (MFA), to automatically detect PSEs. However, MFA's performance was not statistically validated, and it is unclear under what conditions it is valid. This paper's contributions are a power study that validates

the MFA; an alternative test, the single-snippet analysis (SSA), which has the same functionality as MFA but is easier to calculate and has better power in small samples; a simple bootstrap simulation to estimate SpTA baselines with simulation bands that help visualize potential PSEs; and a bootstrap adjustment to the MFA and SSA to correct for nonlinear SpTA baselines.

---

4. ParceLiNGAM: A Causal Ordering Method Robust Against Latent Confounders

Tatsuya Tashiro, Shohei Shimizu, Aapo Hyvärinen, Takashi Washio

https://doi.org/10.1162/neco_a_00533 | pp. 57-83

We consider learning a causal ordering of variables in a linear nongaussian acyclic model called LiNGAM. Several methods have been shown to consistently estimate a causal ordering assuming that all the model assumptions are correct. But the estimation results could be distorted if some assumptions are violated. In this letter, we propose a new algorithm for learning causal orders that is robust against one typical violation of the model assumptions: latent confounders. The key idea is to detect latent confounders by testing independence between estimated external influences and find subsets (parcels) that include variables unaffected by latent confounders. We demonstrate the effectiveness of our method using artificial data and simulated brain imaging data.

---

5. Information-Maximization Clustering Based on Squared-Loss Mutual Information

Masashi Sugiyama, Gang Niu, Makoto Yamada, Manabu Kimura, Hirotaka Hachiya

https://doi.org/10.1162/neco_a_00534 | pp. 84-131

Information-maximization clustering learns a probabilistic classifier in an unsupervised manner so that mutual information between feature vectors and cluster assignments is maximized. A notable advantage of this approach is that it involves only continuous optimization of model parameters, which is substantially simpler than discrete optimization of cluster assignments. However, existing methods still involve nonconvex optimization problems, and therefore finding a good local optimal solution is not straightforward in practice. In this letter, we propose an alternative information-maximization clustering method based on a squared-loss variant of mutual information. This novel approach gives a clustering solution analytically in a computationally efficient way via kernel eigenvalue decomposition. Furthermore, we provide a practical model selection procedure that allows us to objectively optimize tuning parameters included in the kernel function. Through experiments, we demonstrate the usefulness of the proposed approach.

---

6. Local Linear Approximation of the Jacobian Matrix Better Captures Phase Resetting of Neural Limit Cycle Oscillators

Sorinel A. Oprisan

https://doi.org/10.1162/neco_a_00536 | pp. 132-157

One effect of any external perturbations, such as presynaptic inputs, received by limit cycle oscillators when they are part of larger neural networks is a transient change in their firing rate, or phase resetting. A brief

external perturbation moves the figurative point outside the limit cycle, a geometric perturbation that we mapped into a transient change in the firing rate, or a temporal phase resetting. In order to gain a better qualitative understanding of the link between the geometry of the limit cycle and the phase resetting curve (PRC), we used a moving reference frame with one axis tangent and the others normal to the limit cycle. We found that the stability coefficients associated with the unperturbed limit cycle provided good quantitative predictions of both the tangent and the normal geometric displacements induced by external perturbations. A geometric-to-temporal mapping allowed us to correctly predict the PRC while preserving the intuitive nature of this geometric approach.

---

7. Learning with Convex Loss and Indefinite Kernels

Hongzhi Tong, Di■Rong Chen, Fenghong Yang

https://doi.org/10.1162/neco_a_00535 | pp. 158-184

We consider a kind of kernel-based regression with general convex loss functions in a regularization scheme. The kernels used in the scheme are not necessarily symmetric and thus are not positive semidefinite; l(1)-norm of the coefficients in the kernel ensembles is taken as the regularizer. Our setting in this letter is quite different from the classical regularized regression algorithms such as regularized networks and support vector machines regression. Under an established error decomposition that consists of approximation error, hypothesis error, and sample error, we present a detailed mathematical analysis for this scheme and, in particular, its learning rate. A reweighted empirical process theory is applied to the analysis of produced learning algorithms, which plays a key role in deriving the explicit learning rate under some assumptions.

---

8. High-Dimensional Feature Selection by Feature-Wise Kernelized Lasso

Makoto Yamada, Wittawat Jitkrittum, Leonid Sigal, Eric P. Xing, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00537 | pp. 185-207

The goal of supervised feature selection is to find a subset of input features that are responsible for predicting output values. The least absolute shrinkage and selection operator (Lasso) allows computationally efficient feature selection based on linear dependency between input features and output values. In this letter, we consider a feature-wise kernelized Lasso for capturing nonlinear input-output dependency. We first show that with particular choices of kernel functions, nonredundant features with strong statistical dependence on output values can be found in terms of kernel-based independence measures such as the Hilbert-Schmidt independence criterion. We then show that the globally optimal solution can be efficiently computed; this makes the approach scalable to high-dimensional problems. The effectiveness of the proposed method is demonstrated through feature selection experiments for classification and regression with thousands of features.

---

9. Blocked 3×2 Cross-Validated t-Test for Comparing Supervised Classification Learning Algorithms

Yu Wang, Wang Ruibo, Jia Huichen, Jihong Li

https://doi.org/10.1162/neco_a_00532 | pp. 208-235

In the research of machine learning algorithms for classification tasks, the comparison of the performances of algorithms is extremely important, and a statistical test of significance for generalization error is often used to perform it in the machine learning literature. In view of the randomness of partitions in cross-validation, a new blocked 3×2 cross-validation is proposed to estimate generalization error in this letter. We then conduct an analysis of variance of the blocked 3×2 cross-validated estimator. A relatively conservative variance estimator that considers the correlation between any two two-fold cross-validations, and was previously neglected in 5×2 cross-validated t and F-tests is put forward. A corresponding test using this variance estimator is presented to compare the performances of algorithms. Simulated results show that the performance of our test is comparable with that of 5×2 cross-validated tests but with less computation complexity.

---

10. Likelihood Methods for Point Processes with Refractoriness

Luca Citi, Demba Ba, Emery N. Brown, Riccardo Barbieri

https://doi.org/10.1162/neco_a_00548 | pp. 237-263

Likelihood-based encoding models founded on point processes have received significant attention in the literature because of their ability to reveal the information encoded by spiking neural populations. We propose an approximation to the likelihood of a point-process model of neurons that holds under assumptions about the continuous time process that are physiologically reasonable for neural spike trains: the presence of a refractory period, the predictability of the conditional intensity function, and its integrability. These are properties that apply to a large class of point processes arising in applications other than neuroscience. The proposed approach has several advantages over conventional ones. In particular, one can use standard fitting procedures for generalized linear models based on iteratively reweighted least squares while improving the accuracy of the approximation to the likelihood and reducing bias in the estimation of the parameters of the underlying continuous-time model. As a result, the proposed approach can use a larger bin size to achieve the same accuracy as conventional approaches would with a smaller bin size. This is particularly important when analyzing neural data with high mean and instantaneous firing rates. We demonstrate these claims on simulated and real neural spiking activity. By allowing a substantive increase in the required bin size, our algorithm has the potential to lower the barrier to the use of point-process methods in an increasing number of applications.

---

11. Functional Identification of Spike-Processing Neural Circuits

Aurel A. Lazar, Yevgeniy B. Slutskiy

https://doi.org/10.1162/neco_a_00543 | pp. 264-305

We introduce a novel approach for a complete functional identification of biophysical spike-processing neural circuits. The circuits considered accept multidimensional spike trains as their input and comprise a multitude of temporal receptive fields and conductance-based models of action potential generation. Each temporal receptive field describes the spatiotemporal contribution of all synapses between any two neurons and incorporates the (passive) processing carried out by the dendritic tree. The aggregate dendritic current produced by a multitude of temporal receptive fields is encoded into a sequence of action potentials by a spike generator modeled as a nonlinear dynamical system. Our approach builds on the observation that

during any experiment, an entire neural circuit, including its receptive fields and biophysical spike generators, is projected onto the space of stimuli used to identify the circuit. Employing the reproducing kernel Hilbert space (RKHS) of trigonometric polynomials to describe input stimuli, we quantitatively describe the relationship between underlying circuit parameters and their projections. We also derive experimental conditions under which these projections converge to the true parameters. In doing so, we achieve the mathematical tractability needed to characterize the biophysical spike generator and identify the multitude of receptive fields. The algorithms obviate the need to repeat experiments in order to compute the neurons' rate of response, rendering our methodology of interest to both experimental and theoretical neuroscientists.

---

12. A New Class of Metrics for Spike Trains

C■t■lin V. Rusu, R■zvan V. Florian

https://doi.org/10.1162/neco_a_00545 | pp. 306-348

The distance between a pair of spike trains, quantifying the differences between them, can be measured using various metrics. Here we introduce a new class of spike train metrics, inspired by the Pompeiu-Hausdorff distance, and compare them with existing metrics. Some of our new metrics (the modulus-metric and the max-metric) have characteristics that are qualitatively different from those of classical metrics like the van Rossum distance or the Victor and Purpura distance. The modulus-metric and the max-metric are particularly suitable for measuring distances between spike trains where information is encoded in bursts, but the number and the timing of spikes inside a burst do not carry information. The modulus-metric does not depend on any parameters and can be computed using a fast algorithm whose time depends linearly on the number of spikes in the two spike trains. We also introduce localized versions of the new metrics, which could have the biologically relevant interpretation of measuring the differences between spike trains as they are perceived at a particular moment in time by a neuron receiving these spike trains.

---

13. Robust Common Spatial Filters with a Maxmin Approach

Motoaki Kawanabe, Wojciech Samek, Klaus■Robert Müller, Carmen Vidaurre

https://doi.org/10.1162/neco_a_00544 | pp. 349-376

Electroencephalographic signals are known to be nonstationary and easily affected by artifacts; therefore, their analysis requires methods that can deal with noise. In this work, we present a way to robustify the popular common spatial patterns (CSP) algorithm under a maxmin approach. In contrast to standard CSP that maximizes the variance ratio between two conditions based on a single estimate of the class covariance matrices, we propose to robustly compute spatial filters by maximizing the minimum variance ratio within a prefixed set of covariance matrices called the tolerance set. We show that this kind of maxmin optimization makes CSP robust to outliers and reduces its tendency to overfit. We also present a data-driven approach to construct a tolerance set that captures the variability of the covariance matrices over time and shows its ability to reduce the nonstationarity of the extracted features and significantly improve classification accuracy. We test the spatial filters derived with this approach and compare them to standard CSP and a state-of-the-art method on a real-world brain-computer interface (BCI) data set in which we expect substantial fluctuations caused by environmental differences. Finally we investigate the advantages and

limitations of the maxmin approach with simulations.

---

14. Improved Sparse Coding Under the Influence of Perceptual Attention

Ashkan Amiri, S. Haykin

https://doi.org/10.1162/neco_a_00546 | pp. 377-420

Sparse coding has established itself as a useful tool for the representation of natural data in the neuroscience as well as signal-processing literature. The aim of this letter, inspired by the human brain, is to improve on the performance of the sparse coding algorithm by trying to bridge the gap between neuroscience and engineering. To this end, we build on the localized perception-action cycle in cognitive neuroscience by categorizing it under the umbrella of perceptual attention, which lends itself to increase gradually the contrast between relevant information and irrelevant information. Stated in another way, irrelevant information is filtered away, while relevant information about the environment is enhanced from one cycle to the next. We may thus think in terms of the information filter, which, in a Bayesian context, was introduced in the literature by Fraser (1967). In a Bayesian context, the information filter provides a method for algorithmic implementation of perceptual attention. The information filter may therefore be viewed as the basis for improving the algorithmic performance of sparse coding. To support this performance improvement, the letter presents two computer experiments. The first experiment uses simulated (real-valued) data that are generated to purposely make the problem challenging. The second uses real-life radar data that are complex valued, hence the proposal to introduce Wirtinger calculus into derivation of the new algorithm.

---

15. Spontaneous Clustering via Minimum Gamma-Divergence

Akifumi Notsu, Osamu Komori, Shinto Eguchi

https://doi.org/10.1162/neco_a_00547 | pp. 421-448

We propose a new method for clustering based on local minimization of the gamma-divergence, which we call spontaneous clustering. The greatest advantage of the proposed method is that it automatically detects the number of clusters that adequately reflect the data structure. In contrast, existing methods, such as K-means, fuzzy c-means, or model-based clustering need to prescribe the number of clusters. We detect all the local minimum points of the gamma-divergence, by which we define the cluster centers. A necessary and sufficient condition for the gamma-divergence to have local minimum points is also derived in a simple setting. Applications to simulated and real data are presented to compare the proposed method with existing ones.

---

16. A Novel Iterative Method for Computing Generalized Inverse

Youshen Xia, Tianping Chen, Jinjun Shan

https://doi.org/10.1162/neco_a_00549 | pp. 449-465

In this letter, we propose a novel iterative method for computing generalized inverse, based on a novel KKT formulation. The proposed iterative algorithm requires making four matrix and vector multiplications at each iteration and thus has low computational complexity. The proposed method is proved to be globally

convergent without any condition. Furthermore, for fast computing generalized inverse, we present an acceleration scheme based on the proposed iterative method. The global convergence of the proposed acceleration algorithm is also proved. Finally, the effectiveness of the proposed iterative algorithm is evaluated numerically.

---

17. Dopamine Ramps Are a Consequence of Reward Prediction Errors

Samuel J. Gershman

https://doi.org/10.1162/neco_a_00559 | pp. 467-471

Temporal difference learning models of dopamine assert that phasic levels of dopamine encode a reward prediction error. However, this hypothesis has been challenged by recent observations of gradually ramping stratal dopamine levels as a goal is approached. This note describes conditions under which temporal difference learning models predict dopamine ramping. The key idea is representational: a quadratic transformation of proximity to the goal implies approximately linear ramping, as observed experimentally.

---

18. Approximate, Computationally Efficient Online Learning in Bayesian Spiking Neurons

Levin Kuhlmann, Michael Hauser-Raspe, Jonathan H. Manton, David B. Grayden, Jonathan Tapson, André van Schaik

https://doi.org/10.1162/neco_a_00560 | pp. 472-496

Bayesian spiking neurons (BSNs) provide a probabilistic interpretation of how neurons perform inference and learning. Online learning in BSNs typically involves parameter estimation based on maximum-likelihood expectation-maximization (ML-EM) which is computationally slow and limits the potential of studying networks of BSNs. An online learning algorithm, fast learning (FL), is presented that is more computationally efficient than the benchmark ML-EM for a fixed number of time steps as the number of inputs to a BSN increases (e.g., 16.5 times faster run times for 20 inputs). Although ML-EM appears to converge 2.0 to 3.6 times faster than FL, the computational cost of ML-EM means that ML-EM takes longer to simulate to convergence than FL. FL also provides reasonable convergence performance that is robust to initialization of parameter estimates that are far from the true parameter values. However, parameter estimation depends on the range of true parameter values. Nevertheless, for a physiologically meaningful range of parameter values, FL gives very good average estimation accuracy, despite its approximate nature. The FL algorithm therefore provides an efficient tool, complementary to ML-EM, for exploring BSN networks in more detail in order to better understand their biological relevance. Moreover, the simplicity of the FL algorithm means it can be easily implemented in neuromorphic VLSI such that one can take advantage of the energy-efficient spike coding of BSNs.

---

19. Guaranteed Classification via Regularized Similarity Learning

Zheng-Chu Guo, Yiming Ying

https://doi.org/10.1162/neco_a_00556 | pp. 497-522

Learning an appropriate (dis)similarity function from the available data is a central problem in machine learning, since the success of many machine learning algorithms critically depends on the choice of a similarity function to compare examples. Despite many approaches to similarity metric learning that have been proposed, there has been little theoretical study on the links between similarity metric learning and the classification performance of the resulting classifier. In this letter, we propose a regularized similarity learning formulation associated with general matrix norms and establish their generalization bounds. We show that the generalization error of the resulting linear classifier can be bounded by the derived generalization bound of similarity learning. This shows that a good generalization of the learned similarity function guarantees a good classification of the resulting linear classifier. Our results extend and improve those obtained by Bellet, Habrard, and Sebban (2012). Due to the techniques dependent on the notion of uniform stability (Bousquet & Elisseeff, 2002), the bound obtained there holds true only for the Frobenius matrix-norm regularization. Our techniques using the Rademacher complexity (Bartlett & Mendelson, 2002) and its related Khinchin-type inequality enable us to establish bounds for regularized similarity learning formulations associated with general matrix norms, including sparse L1-norm and mixed (2,1)-norm.

---

20. Noise-Robust Speech Recognition Through Auditory Feature Detection and Spike Sequence Decoding

Phillip B. Schafer, Dezhe Z. Jin

https://doi.org/10.1162/neco_a_00557 | pp. 523-556

Speech recognition in noisy conditions is a major challenge for computer systems, but the human brain performs it routinely and accurately. Automatic speech recognition (ASR) systems that are inspired by neuroscience can potentially bridge the performance gap between humans and machines. We present a system for noise-robust isolated word recognition that works by decoding sequences of spikes from a population of simulated auditory feature-detecting neurons. Each neuron is trained to respond selectively to a brief spectrotemporal pattern, or feature, drawn from the simulated auditory nerve response to speech. The neural population conveys the time-dependent structure of a sound by its sequence of spikes. We compare two methods for decoding the spike sequences--one using a hidden Markov model-based recognizer, the other using a novel template-based recognition scheme. In the latter case, words are recognized by comparing their spike sequences to template sequences obtained from clean training data, using a similarity measure based on the length of the longest common sub-sequence. Using isolated spoken digits from the AURORA-2 database, we show that our combined system outperforms a state-of-the-art robust speech recognizer at low signal-to-noise ratios. Both the spike-based encoding scheme and the template-based decoding offer gains in noise robustness over traditional speech recognition methods. Our system highlights potential advantages of spike-based acoustic coding and provides a biologically motivated framework for robust ASR development.

---

21. Feature Selection for Ordinal Text Classification

Stefano Baccianella, Andrea Esuli, Fabrizio Sebastiani

https://doi.org/10.1162/neco_a_00558 | pp. 557-591

Ordinal classification (also known as ordinal regression) is a supervised learning task that consists of estimating the rating of a data item on a fixed, discrete rating scale. This problem is receiving increased

attention from the sentiment analysis and opinion mining community due to the importance of automatically rating large amounts of product review data in digital form. As in other supervised learning tasks such as binary or multiclass classification, feature selection is often needed in order to improve efficiency and avoid overfitting. However, although feature selection has been extensively studied for other classification tasks, it has not for ordinal classification. In this letter, we present six novel feature selection methods that we have specifically devised for ordinal classification and test them on two data sets of product review data against three methods previously known from the literature, using two learning algorithms from the support vector regression tradition. The experimental results show that all six proposed metrics largely outperform all three baseline techniques (and are more stable than these others by an order of magnitude), on both data sets and for both learning algorithms.

---

22. A Parallel Dual Matrix Method for Blind Signal Separation

Tiaojun Zeng, Quan Yuan Feng

https://doi.org/10.1162/neco_a_00554 | pp. 592-610

A parallel dual matrix method that considers all cases of numerical relations between a mixing matrix and a separating matrix is proposed in this letter. Different constrained terms are used to construct cost function for every subalgorithm. These constrained terms reflect numerical relation. Therefore, a number of undesired solutions are excluded, the search region is reduced, and the convergence efficiency of the algorithm is ultimately improved. Moreover, any parallel subalgorithm is proven to converge to a desired separating matrix only if its cost function converges to zero. Computer simulations indicate that the algorithm efficiently performs blind signal separation.

---

23. Robust Subspace Discovery via Relaxed Rank Minimization

Xinggang Wang, Zhengdong Zhang, Yi Ma, Xiang Bai, Wenyu Liu, Zhuowen Tu

https://doi.org/10.1162/neco_a_00555 | pp. 611-635

This letter examines the problem of robust subspace discovery from input data samples (instances) in the presence of overwhelming outliers and corruptions. A typical example is the case where we are given a set of images; each image contains, for example, a face at an unknown location of an unknown size; our goal is to identify or detect the face in the image and simultaneously learn its model. We employ a simple generative subspace model and propose a new formulation to simultaneously infer the label information and learn the model using low-rank optimization. Solving this problem enables us to simultaneously identify the ownership of instances to the subspace and learn the corresponding subspace model. We give an efficient and effective algorithm based on the alternating direction method of multipliers and provide extensive simulations and experiments to verify the effectiveness of our method. The proposed scheme can also be used to tackle many related high-dimensional combinatorial selection problems.

---

24. Optimality and Saturation in Axonal Chemotaxis

Jiajia Yuan, Stanley M. H. Chan, Duncan Mortimer, Huyen Nguyen, Geoffrey J. Goodhill

https://doi.org/10.1162/neco_a_00426 | pp. 833-853

Chemotaxis (detecting and following chemical gradients) plays a crucial role in the function of many biological systems. In particular, gradient following by neuronal growth cones is important for the correct wiring of the nervous system. There is increasing interest in the constraints that determine how small chemotacting devices respond to gradients, but little quantitative information is available in this regard for neuronal growth cones. Mortimer et al. (2009) and Mortimer, Dayan, Burrage, and Goodhill (2011) proposed a Bayesian ideal observer model that predicts chemotactic performance for shallow gradients. Here we investigated two important aspects of this model. First, we found by numerical simulation that although the analytical predictions of the model assume shallow gradients, these predictions remain remarkably robust to large deviations in gradient steepness. Second, we found experimentally that the chemotactic response increased linearly with gradient steepness for very shallow gradients as predicted by the model; however, the response saturated for steeper gradients. This saturation could be reproduced in simulations of a growth rate modulation response mechanism. Together these results illuminate the domain of validity of the Bayesian model and give further insight into the biological mechanisms of axonal chemotaxis.

---

25. Information Transmission Using Non-Poisson Regular Firing

Shinsuke Koyama, Takahiro Omi, Robert E. Kass, Shigeru Shinomoto

https://doi.org/10.1162/neco_a_00420 | pp. 854-876

In many cortical areas, neural spike trains do not follow a Poisson process. In this study, we investigate a possible benefit of non-Poisson spiking for information transmission by studying the minimal rate fluctuation that can be detected by a Bayesian estimator. The idea is that an inhomogeneous Poisson process may make it difficult for downstream decoders to resolve subtle changes in rate fluctuation, but by using a more regular non-Poisson process, the nervous system can make rate fluctuations easier to detect. We evaluate the degree to which regular firing reduces the rate fluctuation detection threshold. We find that the threshold for detection is reduced in proportion to the coefficient of variation of interspike intervals.

---

26. Continuation-Based Numerical Detection of After-Depolarization and Spike-Adding Thresholds

Jakub Nowacki, Hinke M. Osinga, Krasimira Tsaneva■Atanasova

https://doi.org/10.1162/neco_a_00425 | pp. 877-900

The changes in neuronal firing pattern are signatures of brain function, and it is of interest to understand how such changes evolve as a function of neuronal biophysical properties. We address this important problem by the analysis and numerical investigation of a class of mechanistic mathematical models. We focus on a hippocampal pyramidal neuron model and study the occurrence of bursting related to the after-depolarization (ADP) that follows a brief current injection. This type of burst is a transient phenomenon that is not amenable to the classical bifurcation analysis done, for example, for periodic bursting oscillators. In this letter, we show how to formulate such transient behavior as a two-point boundary value problem (2PBVP), which can be solved using well-known continuation methods. The 2PBVP is formulated such that the transient response is represented by a finite orbit segment for which onsets of ADP and additional spikes in a burst can be detected as bifurcations during a one-parameter continuation. This in turn provides us with a direct method to approximate the boundaries of regions in a two-parameter plane where certain model

behavior of interest occurs. More precisely, we use two-parameter continuation of the detected onset points to identify the boundaries between regions with and without ADP and bursts with different numbers of spikes. Our 2PBVP formulation is a novel approach to parameter sensitivity analysis that can be applied to a wide range of problems.

---

27. Some Sampling Properties of Common Phase Estimators

Kyle Q. Lepage, Mark Kramer, Uri T. Eden

https://doi.org/10.1162/neco_a_00422 | pp. 901-921

The instantaneous phase of neural rhythms is important to many neuroscience-related studies. In this letter, we show that the statistical sampling properties of three instantaneous phase estimators commonly employed to analyze neuroscience data share common features, allowing an analytical investigation into their behavior. These three phase estimators-the Hilbert, complex Morlet, and discrete Fourier transform-are each shown to maximize the likelihood of the data, assuming the observation of different neural signals. This connection, explored with the use of a geometric argument, is used to describe the bias and variance properties of each of the phase estimators, their temporal dependence, and the effect of model misspecification. This analysis suggests how prior knowledge about a rhythmic signal can be used to improve the accuracy of phase estimates.

---

28. Double-Gabor Filters Are Independent Components of Small Translation-Invariant Image Patches

Saeed Saremi, Terrence J. Sejnowski, Tatyana O. Sharpee

https://doi.org/10.1162/neco_a_00418 | pp. 922-939

The analysis of natural images with independent component analysis (ICA) yields localized bandpass Gabor-type filters similar to receptive fields of simple cells in visual cortex. We applied ICA on a subset of patches called position-centered patches, selected for forming a translation-invariant representation of small patches. The resulting filters were qualitatively different in two respects. One novel feature was the emergence of filters we call double-Gabor filters. In contrast to Gabor functions that are modulated in one direction, double-Gabor filters are sinusoidally modulated in two orthogonal directions. In addition the filters were more extended in space and frequency compared to standard ICA filters and better matched the distribution in experimental recordings from neurons in primary visual cortex. We further found a dual role for double-Gabor filters as edge and texture detectors, which could have engineering applications.

---

29. Solving the Distal Reward Problem with Rare Correlations

Andrea Soltoggio, Jochen J. Steil

https://doi.org/10.1162/neco_a_00419 | pp. 940-978

In the course of trial-and-error learning, the results of actions, manifested as rewards or punishments, occur often seconds after the actions that caused them. How can a reward be associated with an earlier action when the neural activity that caused that action is no longer present in the network? This problem is referred to as the distal reward problem. A recent computational study proposes a solution using modulated plasticity

with spiking neurons and argues that precise firing patterns in the millisecond range are essential for such a solution. In contrast, the study reported in this letter shows that it is the rarity of correlating neural activity, and not the spike timing, that allows the network to solve the distal reward problem. In this study, rare correlations are detected in a standard rate-based computational model by means of a threshold-augmented Hebbian rule. The novel modulated plasticity rule allows a randomly connected network to learn in classical and instrumental conditioning scenarios with delayed rewards. The rarity of correlations is shown to be a pivotal factor in the learning and in handling various delays of the reward. This study additionally suggests the hypothesis that short-term synaptic plasticity may implement eligibility traces and thereby serve as a selection mechanism in promoting candidate synapses for long-term storage.

---

30. Modular Encoding and Decoding Models Derived from Bayesian Canonical Correlation Analysis

Yusuke Fujiwara, Yoichi Miyawaki, Yukiyasu Kamitani

https://doi.org/10.1162/neco_a_00423 | pp. 979-1005

Neural encoding and decoding provide perspectives for understanding neural representations of sensory inputs. Recent functional magnetic resonance imaging (fMRI) studies have succeeded in building prediction models for encoding and decoding numerous stimuli by representing a complex stimulus as a combination of simple elements. While arbitrary visual images were reconstructed using a modular model that combined the outputs of decoder modules for multiscale local image bases (elements), the shapes of the image bases were heuristically determined. In this work, we propose a method to establish mappings between the stimulus and the brain by automatically extracting modules from measured data. We develop a model based on Bayesian canonical correlation analysis, in which each module is modeled by a latent variable that relates a set of pixels in a visual image to a set of voxels in an fMRI activity pattern. The estimated mapping from a latent variable to pixels can be regarded as an image basis. We show that the model estimates a modular representation with spatially localized multiscale image bases. Further, using the estimated mappings, we derive encoding and decoding models that produce accurate predictions for brain activity and stimulus images. Our approach thus provides a novel means of revealing neural representations of stimuli by automatically extracting modules, which can be used to generate effective prediction models for encoding and decoding.

---

31. A Self-Organized Neural Comparator

Guillermo A. Ludueña, Claudius Gros

https://doi.org/10.1162/neco_a_00424 | pp. 1006-1028

Learning algorithms need generally the ability to compare several streams of information. Neural learning architectures hence need a unit, a comparator, able to compare several inputs encoding either internal or external information, for instance, predictions and sensory readings. Without the possibility of comparing the values of predictions to actual sensory inputs, reward evaluation and supervised learning would not be possible. Comparators are usually not implemented explicitly. Necessary comparisons are commonly performed by directly comparing the respective activities one-to-one. This implies that the characteristics of the two input streams (like size and encoding) must be provided at the time of designing the system. It is, however, plausible that biological comparators emerge from self-organizing, genetically encoded principles,

which allow the system to adapt to the changes in the input and the organism. We propose an unsupervised neural circuitry, where the function of input comparison emerges via self-organization only from the interaction of the system with the respective inputs, without external influence or supervision. The proposed neural comparator adapts in an unsupervised form according to the correlations present in the input streams. The system consists of a multilayer feedforward neural network, which follows a local output minimization (anti-Hebbian) rule for adaptation of the synaptic weights. The local output minimization allows the circuit to autonomously acquire the capability of comparing the neural activities received from different neural populations, which may differ in population size and the neural encoding used. The comparator is able to compare objects never encountered before in the sensory input streams and evaluate a measure of their similarity even when differently encoded.

---

32. Learning with Boundary Conditions

Giorgio Gnecco, Marco Gori, Marcello Sanguineti

https://doi.org/10.1162/neco_a_00417 | pp. 1029-1106

Kernel machines traditionally arise from an elegant formulation based on measuring the smoothness of the admissible solutions by the norm in the reproducing kernel Hilbert space (RKHS) generated by the chosen kernel. It was pointed out that they can be formulated in a related functional framework, in which the Green's function of suitable differential operators is thought of as a kernel. In this letter, our own picture of this intriguing connection is given by emphasizing some relevant distinctions between these different ways of measuring the smoothness of admissible solutions. In particular, we show that for some kernels, there is no associated differential operator. The crucial relevance of boundary conditions is especially emphasized, which is in fact the truly distinguishing feature of the approach based on differential operators. We provide a general solution to the problem of learning from data and boundary conditions and illustrate the significant role played by boundary conditions with examples. It turns out that the degree of freedom that arises in the traditional formulation of kernel machines is indeed a limitation, which is partly overcome when incorporating the boundary conditions. This likely holds true in many real-world applications in which there is prior knowledge about the expected behavior of classifiers and regressors on the boundary.

---

33. Error Analysis of Coefficient-Based Regularized Algorithm for Density-Level Detection

Hong Chen, Zhibin Pan, Luoqing Li, Yuanyan Tang

https://doi.org/10.1162/neco_a_00421 | pp. 1107-1121

In this letter, we consider a density-level detection (DLD) problem by a coefficient-based classification framework with [Formula: see text]-regularizer and data-dependent hypothesis spaces. Although the data-dependent characteristic of the algorithm provides flexibility and adaptivity for DLD, it leads to difficulty in generalization error analysis. To overcome this difficulty, an error decomposition is introduced from an established classification framework. On the basis of this decomposition, the estimate of the learning rate is obtained by using Rademacher average and stepping-stone techniques. In particular, the estimate is independent of the capacity assumption used in the previous literature.

---

34. Impact of Spike Train Autostructure on Probability Distribution of Joint Spike Events

Gordon Pipa, Sonja Grün, Carl van Vreeswijk

https://doi.org/10.1162/neco_a_00432 | pp. 1123-1163

The discussion whether temporally coordinated spiking activity really exists and whether it is relevant has been heated over the past few years. To investigate this issue, several approaches have been taken to determine whether synchronized events occur significantly above chance, that is, whether they occur more often than expected if the neurons fire independently. Most investigations ignore or destroy the autostructure of the spiking activity of individual cells or assume Poissonian spiking as a model. Such methods that ignore the autostructure can significantly bias the coincidence statistics. Here, we study the influence of the autostructure on the probability distribution of coincident spiking events between tuples of mutually independent non-Poisson renewal processes. In particular, we consider two types of renewal processes that were suggested as appropriate models of experimental spike trains: a gamma and a log-normal process. For a gamma process, we characterize the shape of the distribution analytically with the Fano factor (FFc). In addition, we perform Monte Carlo estimations to derive the full shape of the distribution and the probability for false positives if a different process type is assumed as was actually present. We also determine how manipulations of such spike trains, here dithering, used for the generation of surrogate data change the distribution of coincident events and influence the significance estimation. We find, first, that the width of the coincidence count distribution and its FFc depend critically and in a nontrivial way on the detailed properties of the structure of the spike trains as characterized by the coefficient of variation CV. Second, the dependence of the FFc on the CV is complex and mostly nonmonotonic. Third, spike dithering, even if as small as a fraction of the interspike interval, can falsify the inference on coordinated firing.

---

35. Ambient GABA Responsible for Age-Related Changes in Multistable Perception

Osamu Hoshino

https://doi.org/10.1162/neco_a_00431 | pp. 1164-1190

Multistable perception is a psychophysical phenomenon in which one unique interpretation alternates spontaneously every few seconds between two or more interpretations of the same sensory input. Well-known examples include the Necker cube and face-vase illusions in vision. Interestingly, young adults generally see more perceptual switches than do elderly people. To understand the underlying neuronal mechanism of age-related multistable perception, we simulated a cortical neural network model that consists of multiple cell assemblies. In the network, a specific population of noncore cells and a common population of core cells form a cell assembly that represents a single object (or event). Every dynamic cell assembly, activated by a given sensory input, involves the common (overlapping) population of core cells. Ambient GABA-mediated intracortical tonic inhibition via extrasynaptic GABAa receptors destabilized the currently appearing dynamic cell assembly and terminated its burst firing. This allowed another dynamic cell assembly to emerge one after the other. Namely, multistable perception took place. Transporters, which were embedded in axon terminal membranes of interneurons, regulated levels of ambient GABA. For elderly people, we assumed a decline in transporter. This decelerated GABA augmentation and resulted in prolonging the durations of burst firing and thus in slowing perceptual switches. We suggest that poor control of ambient GABA levels due to age-related decline in GABA transporter may be responsible for the slowing

of perceptual switches in elderly people.

---

36. A Detailed and Fast Model of Extracellular Recordings

Luis A. Camuñas-Mesa, Rodrigo Quian Quiroga

https://doi.org/10.1162/neco_a_00433 | pp. 1191-1212

We present a novel method to generate realistic simulations of extracellular recordings. The simulations were obtained by superimposing the activity of neurons placed randomly in a cube of brain tissue. Detailed models of individual neurons were used to reproduce the extracellular action potentials of close-by neurons. To reduce the computational load, the contributions of neurons further away were simulated using previously recorded spikes with their amplitude normalized by the distance to the recording electrode. For making the simulations more realistic, we also considered a model of a finite-size electrode by averaging the potential along the electrode surface and modeling the electrode-tissue interface with a capacitive filter. This model allowed studying the effect of the electrode diameter on the quality of the recordings and how it affects the number of identified neurons after spike sorting. Given that not all neurons are active at a time, we also generated simulations with different ratios of active neurons and estimated the ratio that matches the signal-to-noise values observed in real data. Finally, we used the model to simulate tetrode recordings.

---

37. Model Reductions for Inference: Generality of Pairwise, Binary, and Planar Factor Graphs

Frederik Eaton, Zoubin Ghahramani

https://doi.org/10.1162/neco_a_00441 | pp. 1213-1260

We offer a solution to the problem of efficiently translating algorithms between different types of discrete statistical model. We investigate the expressive power of three classes of model-those with binary variables, with pairwise factors, and with planar topology-as well as their four intersections. We formalize a notion of "simple reduction" for the problem of inferring marginal probabilities and consider whether it is possible to "simply reduce" marginal inference from general discrete factor graphs to factor graphs in each of these seven subclasses. We characterize the reducibility of each class, showing in particular that the class of binary pairwise factor graphs is able to simply reduce only positive models. We also exhibit a continuous "spectral reduction" based on polynomial interpolation, which overcomes this limitation. Experiments assess the performance of standard approximate inference algorithms on the outputs of our reductions.

---

38. Learning Spatial Invariance with the Trace Rule in Nonuniform Distributions

Jasmin Léveillé, Thomas Hannagan

https://doi.org/10.1162/neco_a_00435 | pp. 1261-1276

Convolutional models of object recognition achieve invariance to spatial transformations largely because of the use of a suitably defined pooling operator. This operator typically takes the form of a max or average function defined across units tuned to the same feature. As a model of the brain's ventral pathway, where computations are carried out by weighted synaptic connections, such pooling can lead to spatial invariance

only if the weights that connect similarly tuned units to a given pooling unit are of approximately equal strengths. How identical weights can be learned in the face of nonuniformly distributed data remains unclear. In this letter, we show how various versions of the trace learning rule can help solve this problem. This allows us in turn to explain previously published results and make recommendations as to the optimal rule for invariance learning.

---

39. Semistability of Steepest Descent with Momentum for Quadratic Functions

Naimin Zhang

https://doi.org/10.1162/neco_a_00436 | pp. 1277-1301

Two steepest-descent algorithms with momentum for quadratic functions are considered. For a given learning rate, the sufficient and necessary conditions for the semistability of the steepest-descent algorithms with momentum are established. Moreover, the optimal momentum factors that generally make for the fastest convergence are obtained.

---

40. A Study on L2-Loss (Squared Hinge-Loss) Multiclass SVM

Ching-pei Lee, Chih■Jen Lin

https://doi.org/10.1162/neco_a_00434 | pp. 1302-1323

Crammer and Singer's method is one of the most popular multiclass support vector machines (SVMs). It considers L1 loss (hinge loss) in a complicated optimization problem. In SVM, squared hinge loss (L2 loss) is a common alternative to L1 loss, but surprisingly we have not seen any paper studying the details of Crammer and Singer's method using L2 loss. In this letter, we conduct a thorough investigation. We show that the derivation is not trivial and has some subtle differences from the L1 case. Details provided in this work can be a useful reference for those who intend to use Crammer and Singer's method with L2 loss. They do not need a tedious process to derive everything by themselves. Furthermore, we present some new results on and discussion of both L1- and L2-loss formulations.

---

41. Relative Density-Ratio Estimation for Robust Distribution Comparison

Makoto Yamada, Taiji Suzuki, Takafumi Kanamori, Hirotaka Hachiya, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00442 | pp. 1324-1370

Divergence estimators based on direct approximation of density ratios without going through separate approximation of numerator and denominator densities have been successfully applied to machine learning tasks that involve distribution comparison such as outlier detection, transfer learning, and two-sample homogeneity test. However, since density-ratio functions often possess high fluctuation, divergence estimation is a challenging task in practice. In this letter, we use relative divergences for distribution comparison, which involves approximation of relative density ratios. Since relative density ratios are always smoother than corresponding ordinary density ratios, our proposed method is favorable in terms of nonparametric convergence speed. Furthermore, we show that the proposed divergence estimator has asymptotic variance independent of the model complexity under a parametric setup, implying that the

proposed estimator hardly overfits even with complex models. Through experiments, we demonstrate the usefulness of the proposed approach.

---

42. Emergence of Optimal Decoding of Population Codes Through STDP

Stefan Habenschuss, Helmut Puhr, Wolfgang Maass

https://doi.org/10.1162/neco_a_00446 | pp. 1371-1407

The brain faces the problem of inferring reliable hidden causes from large populations of noisy neurons, for example, the direction of a moving object from spikes in area MT. It is known that a theoretically optimal likelihood decoding could be carried out by simple linear readout neurons if weights of synaptic connections were set to certain values that depend on the tuning functions of sensory neurons. We show here that such theoretically optimal readout weights emerge autonomously through STDP in conjunction with lateral inhibition between readout neurons. In particular, we identify a class of optimal STDP learning rules with homeostatic plasticity, for which the autonomous emergence of optimal readouts can be explained on the basis of a rigorous learning theory. This theory shows that the network motif we consider approximates expectation-maximization for creating internal generative models for hidden causes of high-dimensional spike inputs. Notably, we find that this optimal functionality can be well approximated by a variety of STDP rules beyond those predicted by theory. Furthermore, we show that this learning process is very stable and automatically adjusts weights to changes in the number of readout neurons, the tuning functions of sensory neurons, and the statistics of external stimuli.

---

43. Randomly Connected Networks Have Short Temporal Memory

Edward W. Wallace, Hamid Reza Maei, Peter E. Latham

https://doi.org/10.1162/neco_a_00449 | pp. 1408-1439

The brain is easily able to process and categorize complex time-varying signals. For example, the two sentences, "It is cold in London this time of year" and "It is hot in London this time of year," have different meanings, even though the words hot and cold appear several seconds before the ends of the two sentences. Any network that can tell these sentences apart must therefore have a long temporal memory. In other words, the current state of the network must depend on events that happened several seconds ago. This is a difficult task, as neurons are dominated by relatively short time constants--tens to hundreds of milliseconds. Nevertheless, it was recently proposed that randomly connected networks could exhibit the long memories necessary for complex temporal processing. This is an attractive idea, both for its simplicity and because little tuning of recurrent synaptic weights is required. However, we show that when connectivity is high, as it is in the mammalian brain, randomly connected networks cannot exhibit temporal memory much longer than the time constants of their constituent neurons.

---

44. New Supervised Learning Theory Applied to Cerebellar Modeling for Suppression of Variability of Saccade End Points

Masahiko Fujita

https://doi.org/10.1162/neco_a_00448 | pp. 1440-1471

A new supervised learning theory is proposed for a hierarchical neural network with a single hidden layer of threshold units, which can approximate any continuous transformation, and applied to a cerebellar function to suppress the end-point variability of saccades. In motor systems, feedback control can reduce noise effects if the noise is added in a pathway from a motor center to a peripheral effector; however, it cannot reduce noise effects if the noise is generated in the motor center itself: a new control scheme is necessary for such noise. The cerebellar cortex is well known as a supervised learning system, and a novel theory of cerebellar cortical function developed in this study can explain the capability of the cerebellum to feedforwardly reduce noise effects, such as end-point variability of saccades. This theory assumes that a Golgi-granule cell system can encode the strength of a mossy fiber input as the state of neuronal activity of parallel fibers. By combining these parallel fiber signals with appropriate connection weights to produce a Purkinje cell output, an arbitrary continuous input-output relationship can be obtained. By incorporating such flexible computation and learning ability in a process of saccadic gain adaptation, a new control scheme in which the cerebellar cortex feedforwardly suppresses the end-point variability when it detects a variation in saccadic commands can be devised. Computer simulation confirmed the efficiency of such learning and showed a reduction in the variability of saccadic end points, similar to results obtained from experimental data.

---

45. A New Supervised Learning Algorithm for Spiking Neurons

Yan Xu, Xiaoqin Zeng, Shuiming Zhong

https://doi.org/10.1162/neco_a_00450 | pp. 1472-1511

The purpose of supervised learning with temporal encoding for spiking neurons is to make the neurons emit a specific spike train encoded by the precise firing times of spikes. If only running time is considered, the supervised learning for a spiking neuron is equivalent to distinguishing the times of desired output spikes and the other time during the running process of the neuron through adjusting synaptic weights, which can be regarded as a classification problem. Based on this idea, this letter proposes a new supervised learning method for spiking neurons with temporal encoding; it first transforms the supervised learning into a classification problem and then solves the problem by using the perceptron learning rule. The experiment results show that the proposed method has higher learning accuracy and efficiency over the existing learning methods, so it is more powerful for solving complex and real-time problems.

---

46. Efficient Sample Reuse in Policy Gradients with Parameter-Based Exploration

Tingting Zhao, Hirotaka Hachiya, Voot Tangkaratt, Jun Morimoto, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00452 | pp. 1512-1547

The policy gradient approach is a flexible and powerful reinforcement learning method particularly for problems with continuous actions such as robot control. A common challenge is how to reduce the variance of policy gradient estimates for reliable policy updates. In this letter, we combine the following three ideas and give a highly effective policy gradient method: (1) policy gradients with parameter-based exploration, a recently proposed policy search method with low variance of gradient estimates; (2) an importance sampling technique, which allows us to reuse previously gathered data in a consistent way; and (3) an optimal baseline, which minimizes the variance of gradient estimates with their unbiasedness being maintained. For

the proposed method, we give a theoretical analysis of the variance of gradient estimates and show its usefulness through extensive experiments.

---

47. The Support Feature Machine: Classification with the Least Number of Features and Application to Neuroimaging Data

Sascha Klement, Silke Anders, Thomas Martinetz

https://doi.org/10.1162/neco_a_00447 | pp. 1548-1584

By minimizing the zero-norm of the separating hyperplane, the support feature machine (SFM) finds the smallest subspace (the least number of features) of a data set such that within this subspace, two classes are linearly separable without error. This way, the dimensionality of the data is more efficiently reduced than with support vector-based feature selection, which can be shown both theoretically and empirically. In this letter, we first provide a new formulation of the previously introduced concept of the SFM. With this new formulation, classification of unbalanced and nonseparable data is straightforward, which allows using the SFM for feature selection and classification in a large variety of different scenarios. To illustrate how the SFM can be used to identify both the smallest subset of discriminative features and the total number of informative features in biological data sets we apply repetitive feature selection based on the SFM to a functional magnetic resonance imaging data set. We suggest that these capabilities qualify the SFM as a universal method for feature selection, especially for high-dimensional small-sample-size data sets that often occur in biological and medical applications.

---

48. Parameter Learning for Alpha Integration

Heeyoul Choi, Seungjin Choi, Yoonsuck Choe

https://doi.org/10.1162/neco_a_00445 | pp. 1585-1604

In pattern recognition, data integration is an important issue, and when properly done, it can lead to improved performance. Also, data integration can be used to help model and understand multimodal processing in the brain. Amari proposed α-integration as a principled way of blending multiple positive measures (e.g., stochastic models in the form of probability distributions), enabling an optimal integration in the sense of minimizing the α-divergence. It also encompasses existing integration methods as its special case, for example, a weighted average and an exponential mixture. The parameter α determines integration characteristics, and the weight vector w assigns the degree of importance to each measure. In most work, however, α and w are given in advance rather than learned. In this letter, we present a parameter learning algorithm for learning α and ω from data when multiple integrated target values are available. Numerical experiments on synthetic as well as real-world data demonstrate the effectiveness of the proposed method.

---

49. Causality in Linear Nongaussian Acyclic Models in the Presence of Latent Gaussian Confounders

Zhitang Chen, Laiwan Chan

https://doi.org/10.1162/neco_a_00444 | pp. 1605-1641

LiNGAM has been successfully applied to some real-world causal discovery problems. Nevertheless, causal sufficiency is assumed; that is, there is no latent confounder of the observations, which may be unrealistic for real-world problems. Taking into the consideration latent confounders will improve the reliability and accuracy of estimations of the real causal structures. In this letter, we investigate a model called linear nongaussian acyclic models in the presence of latent gaussian confounders (LiNGAM-GC) which can be seen as a specific case of lvLiNGAM. This model includes the latent confounders, which are assumed to be independent gaussian distributed and statistically independent of the disturbances. To tackle the causal discovery problem of this model, first we propose a pairwise cumulant-based measure of causal directions for cause-effect pairs. We prove that in spite of the presence of latent gaussian confounders, the causal direction of the observed cause-effect pair can be identified under the mild condition that the disturbances are simultaneously supergaussian or subgaussian. We propose a simple and efficient method to detect the violation of this condition. We extend our work to multivariate causal network discovery problems. Specifically we propose algorithms to estimate the causal network structure, including causal ordering and causal strengths, using an iterative root finding-removing scheme based on pairwise measure. To address the redundant edge problem due to the finite sample size effect, we develop an efficient bootstrapping-based pruning algorithm. Experiments on synthetic data and real-world data have been conducted to show the applicability of our model and the effectiveness of our proposed algorithms.

---

50. Simple Neural-Like P Systems for Maximal Independent Set Selection

Lei Xu, Peter Jeavons

https://doi.org/10.1162/neco_a_00443 | pp. 1642-1659

Membrane systems (P systems) are distributed computing models inspired by living cells where a collection of processors jointly achieves a computing task. The problem of maximal independent set (MIS) selection in a graph is to choose a set of nonadjacent nodes to which no further nodes can be added. In this letter, we design a class of simple neural-like P systems to solve the MIS selection problem efficiently in a distributed way. This new class of systems possesses two features that are attractive for both distributed computing and membrane computing: first, the individual processors do not need any information about the overall size of the graph; second, they communicate using only one-bit messages.

---

51. Learning Quadratic Receptive Fields from Neural Responses to Natural Stimuli

Kanaka Rajan, Olivier Marre, Gašper Tka■ik

https://doi.org/10.1162/neco_a_00463 | pp. 1661-1692

Models of neural responses to stimuli with complex spatiotemporal correlation structure often assume that neurons are selective for only a small number of linear projections of a potentially high-dimensional input. In this review, we explore recent modeling approaches where the neural response depends on the quadratic form of the input rather than on its linear projection, that is, the neuron is sensitive to the local covariance structure of the signal preceding the spike. To infer this quadratic dependence in the presence of arbitrary (e.g., naturalistic) stimulus distribution, we review several inference methods, focusing in particular on two information theory-based approaches (maximization of stimulus energy and of noise entropy) and two likelihood-based approaches (Bayesian spike-triggered covariance and extensions of generalized linear

models). We analyze the formal relationship between the likelihood-based and information-based approaches to demonstrate how they lead to consistent inference. We demonstrate the practical feasibility of these procedures by using model neurons responding to a flickering variance stimulus.

---

52. Design and Analysis of Closed-Loop Decoder Adaptation Algorithms for Brain-Machine Interfaces

Siddharth Dangi, Amy L. Orsborn, Helene G. Moorman, Jose M. Carmena

https://doi.org/10.1162/neco_a_00460 | pp. 1693-1731

Closed-loop decoder adaptation (CLDA) is an emerging paradigm for achieving rapid performance improvements in online brain-machine interface (BMI) operation. Designing an effective CLDA algorithm requires making multiple important decisions, including choosing the timescale of adaptation, selecting which decoder parameters to adapt, crafting the corresponding update rules, and designing CLDA parameters. These design choices, combined with the specific settings of CLDA parameters, will directly affect the algorithm's ability to make decoder parameters converge to values that optimize performance. In this article, we present a general framework for the design and analysis of CLDA algorithms and support our results with experimental data of two monkeys performing a BMI task. First, we analyze and compare existing CLDA algorithms to highlight the importance of four critical design elements: the adaptation timescale, selective parameter adaptation, smooth decoder updates, and intuitive CLDA parameters. Second, we introduce mathematical convergence analysis using measures such as mean-squared error and KL divergence as a useful paradigm for evaluating the convergence properties of a prototype CLDA algorithm before experimental testing. By applying these measures to an existing CLDA algorithm, we demonstrate that our convergence analysis is an effective analytical tool that can ultimately inform and improve the design of CLDA algorithms.

---

53. Decorrelation by Recurrent Inhibition in Heterogeneous Neural Circuits

Alberto Bernacchia, Xiao■Jing Wang

https://doi.org/10.1162/neco_a_00451 | pp. 1732-1767

The activity of neurons is correlated, and this correlation affects how the brain processes information. We study the neural circuit mechanisms of correlations by analyzing a network model characterized by strong and heterogeneous interactions: excitatory input drives the fluctuations of neural activity, which are counterbalanced by inhibitory feedback. In particular, excitatory input tends to correlate neurons, while inhibitory feedback reduces correlations. We demonstrate that heterogeneity of synaptic connections is necessary for this inhibition of correlations. We calculate statistical averages over the disordered synaptic interactions and apply our findings to both a simple linear model and a more realistic spiking network model. We find that correlations at zero time lag are positive and of magnitude K(-1/2) where K is the number of connections to a neuron. Correlations at longer timescales are of smaller magnitude, of order K(-1), implying that inhibition of correlations occurs quickly, on a timescale of K(-1/2). The small magnitude of correlations agrees qualitatively with physiological measurements in the cerebral cortex and basal ganglia. The model could be used to study correlations in brain regions dominated by recurrent inhibition, such as the striatum and globus pallidus.

---

54. Neutral Stability, Rate Propagation, and Critical Branching in Feedforward Networks

N. Alex Cayco-Gajic, Eric Shea■Brown

https://doi.org/10.1162/neco_a_00461 | pp. 1768-1806

Recent experimental and computational evidence suggests that several dynamical properties may characterize the operating point of functioning neural networks: critical branching, neutral stability, and production of a wide range of firing patterns. We seek the simplest setting in which these properties emerge, clarifying their origin and relationship in random, feedforward networks of McCullochs-Pitts neurons. Two key parameters are the thresholds at which neurons fire spikes and the overall level of feedforward connectivity. When neurons have low thresholds, we show that there is always a connectivity for which the properties in question all occur, that is, these networks preserve overall firing rates from layer to layer and produce broad distributions of activity in each layer. This fails to occur, however, when neurons have high thresholds. A key tool in explaining this difference is the eigenstructure of the resulting mean-field Markov chain, as this reveals which activity modes will be preserved from layer to layer. We extend our analysis from purely excitatory networks to more complex models that include inhibition and local noise, and find that both of these features extend the parameter ranges over which networks produce the properties of interest.

---

55. The Role of Extracellular Conductivity Profiles in Compartmental Models for Neurons: Particulars for Layer 5 Pyramidal Cells

Kai Wang, Jorge Riera, Herve Enjieu-Kadji, Ryuta Kawashima

https://doi.org/10.1162/neco_a_00458 | pp. 1807-1852

With the rapid increase in the number of technologies aimed at observing electric activity inside the brain, scientists have felt the urge to create proper links between intracellular- and extracellular-based experimental approaches. Biophysical models at both physical scales have been formalized under assumptions that impede the creation of such links. In this work, we address this issue by proposing a multicompartment model that allows the introduction of complex extracellular and intracellular resistivity profiles. This model accounts for the geometrical and electrotonic properties of any type of neuron through the combination of four devices: the integrator, the propagator, the 3D connector, and the collector. In particular, we applied this framework to model the tufted pyramidal cells of layer 5 (PCL5) in the neocortex. Our model was able to reproduce the decay and delay curves of backpropagating action potentials (APs) in this type of cell with better agreement with experimental data. We used the voltage drops of the extracellular resistances at each compartment to approximate the local field potentials generated by a PCL5 located in close proximity to linear microelectrode arrays. Based on the voltage drops produced by backpropagating APs, we were able to estimate the current multipolar moments generated by a PCL5. By adding external current sources in parallel to the extracellular resistances, we were able to create a sensitivity profile of PCL5 to electric current injections from nearby microelectrodes. In our model for PCL5, the kinetics and spatial profile of each ionic current were determined based on a literature survey, and the geometrical properties of these cells were evaluated experimentally. We concluded that the inclusion of the extracellular space in the compartmental models of neurons as an extra electrotonic medium is crucial for the accurate simulation of both the propagation of the electric potentials along the neuronal dendrites and the neuronal reactivity to an electrical stimulation using external microelectrodes.

---

56. A Calcium-Based Simple Model of Multiple Spike Interactions in Spike-Timing-Dependent Plasticity

Takumi Uramoto, Hiroyuki Torikai

https://doi.org/10.1162/neco_a_00462 | pp. 1853-1869

Spike-timing-dependent plasticity (STDP) is a form of synaptic modification that depends on the relative timings of presynaptic and postsynaptic spikes. In this letter, we proposed a calcium-based simple STDP model, described by an ordinary differential equation having only three state variables: one represents the density of intracellular calcium, one represents a fraction of open state NMDARs, and one represents the synaptic weight. We shown that in spite of its simplicity, the model can reproduce the properties of the plasticity that have been experimentally measured in various brain areas (e.g., layer 2/3 and 5 visual cortical slices, hippocampal cultures, and layer 2/3 somatosensory cortical slices) with respect to various patterns of presynaptic and postsynaptic spikes. In addition, comparisons with other STDP models are made, and the significance and advantages of the proposed model are discussed.

---

57. Identifying Functional Bases for Multidimensional Neural Computations

Joel Thomas Kaardal, Jeffrey D. Fitzgerald, Michael J. Berry, Tatyana O. Sharpee

https://doi.org/10.1162/neco_a_00465 | pp. 1870-1890

Current dimensionality-reduction methods can identify relevant subspaces for neural computations but do not favor one basis over the other within the relevant subspace. Finding the appropriate basis can simplify the description of the nonlinear computation with respect to the relevant variables, making it easier to elucidate the underlying neural computation and make hypotheses about the neural circuitry, giving rise to the observed responses. Part of the problem is that although some of the dimensionality reduction methods can identify many of the relevant dimensions, it is usually difficult to map out or interpret the nonlinear transformation with respect to more than a few relevant dimensions simultaneously without some simplifying assumptions. While recent approaches make it possible to create predictive models based on many relevant dimensions simultaneously, there still remains the need to relate such predictive models to the mechanistic descriptions of the operation of underlying neural circuitry. Here we demonstrate that transforming to a basis within the relevant subspace where the neural computation is best described by a given nonlinear function often makes it easier to interpret the computation and describe it with a small number of parameters. We refer to the corresponding basis as the functional basis, and illustrate the utility of such transformation in the context of logical OR and logical AND functions. We show that although dimensionality-reduction methods such as spike-triggered covariance are able to find a relevant subspace, they often produce dimensions that are difficult to interpret and do not correspond to a functional basis. The functional features can be found using a maximum likelihood approach. The results are illustrated using simulated neurons and recordings from retinal ganglion cells. The resulting features are uniquely defined and nonorthogonal, and they make it easier to relate computational and mechanistic models to each other.

---

58. Combinatorial Neural Codes from a Mathematical Coding Theory Perspective

Carina Curto, Vladimir Itskov, Katherine Morrison, Zachary Roth, Judy L. Walker

https://doi.org/10.1162/neco_a_00459 | pp. 1891-1925

Shannon's seminal 1948 work gave rise to two distinct areas of research: information theory and mathematical coding theory. While information theory has had a strong influence on theoretical neuroscience, ideas from mathematical coding theory have received considerably less attention. Here we take a new look at combinatorial neural codes from a mathematical coding theory perspective, examining the error correction capabilities of familiar receptive field codes (RF codes). We find, perhaps surprisingly, that the high levels of redundancy present in these codes do not support accurate error correction, although the error-correcting performance of receptive field codes catches up to that of random comparison codes when a small tolerance to error is introduced. However, receptive field codes are good at reflecting distances between represented stimuli, while the random comparison codes are not. We suggest that a compromise in error-correcting capability may be a necessary price to pay for a neural code whose structure serves not only error correction, but must also reflect relationships between stimuli.

---

59. Multiple Spectral Kernel Learning and a Gaussian Complexity Computation

Nima Reyhani

https://doi.org/10.1162/neco_a_00457 | pp. 1926-1951

Multiple kernel learning (MKL) partially solves the kernel selection problem in support vector machines and similar classifiers by minimizing the empirical risk over a subset of the linear combination of given kernel matrices. For large sample sets, the size of the kernel matrices becomes a numerical issue. In many cases, the kernel matrix is of low-efficient rank. However, the low-rank property is not efficiently utilized in MKL algorithms. Here, we suggest multiple spectral kernel learning that efficiently uses the low-rank property by finding a kernel matrix from a set of Gram matrices of a few eigenvectors from all given kernel matrices, called a spectral kernel set. We provide a new bound for the gaussian complexity of the proposed kernel set, which depends on both the geometry of the kernel set and the number of Gram matrices. This characterization of the complexity implies that in an MKL setting, adding more kernels may not monotonically increase the complexity, while previous bounds show otherwise.

---

60. Encoding Through Patterns: Regression Tree–Based Neuronal Population Models

Robert Haslinger, Gordon Pipa, Laura D. Lewis, Danko Nikoli■, Ziv M. Williams, Emery N. Brown

https://doi.org/10.1162/neco_a_00464 | pp. 1953-1993

Although the existence of correlated spiking between neurons in a population is well known, the role such correlations play in encoding stimuli is not. We address this question by constructing pattern-based encoding models that describe how time-varying stimulus drive modulates the expression probabilities of population-wide spike patterns. The challenge is that large populations may express an astronomical number of unique patterns, and so fitting a unique encoding model for each individual pattern is not feasible. We avoid this combinatorial problem using a dimensionality-reduction approach based on regression trees. Using the insight that some patterns may, from the perspective of encoding, be statistically indistinguishable, the tree divisively clusters the observed patterns into groups whose member patterns possess similar encoding properties. These groups, corresponding to the leaves of the tree, are much smaller in number than the original patterns, and the tree itself constitutes a tractable encoding model for each pattern. Our

formalism can detect an extremely weak stimulus-driven pattern structure and is based on maximizing the data likelihood, not making a priori assumptions as to how patterns should be grouped. Most important, by comparing pattern encodings with independent neuron encodings, one can determine if neurons in the population are driven independently or collectively. We demonstrate this method using multiple unit recordings from area 17 of anesthetized cat in response to a sinusoidal grating and show that pattern-based encodings are superior to those of independent neuron models. The agnostic nature of our clustering approach allows us to investigate encoding by the collective statistics that are actually present rather than those (such as pairwise) that might be presumed.

---

61. Analysis of the Stabilized Supralinear Network

Yashar Ahmadian, Daniel B. Rubin, Kenneth D. Miller

https://doi.org/10.1162/neco_a_00472 | pp. 1994-2037

We study a rate-model neural network composed of excitatory and inhibitory neurons in which neuronal input-output functions are power laws with a power greater than 1, as observed in primary visual cortex. This supralinear input-output function leads to supralinear summation of network responses to multiple inputs for weak inputs. We show that for stronger inputs, which would drive the excitatory subnetwork to instability, the network will dynamically stabilize provided feedback inhibition is sufficiently strong. For a wide range of network and stimulus parameters, this dynamic stabilization yields a transition from supralinear to sublinear summation of network responses to multiple inputs. We compare this to the dynamic stabilization in the balanced network, which yields only linear behavior. We more exhaustively analyze the two-dimensional case of one excitatory and one inhibitory population. We show that in this case, dynamic stabilization will occur whenever the determinant of the weight matrix is positive and the inhibitory time constant is sufficiently small, and analyze the conditions for supersaturation, or decrease of firing rates with increasing stimulus contrast (which represents increasing input firing rates). In work to be presented elsewhere, we have found that this transition from supralinear to sublinear summation can explain a wide variety of nonlinearities in cerebral cortical processing.

---

62. Representing Objects, Relations, and Sequences

Stephen I. Gallant, T. Wendy Okaywe

https://doi.org/10.1162/neco_a_00467 | pp. 2038-2078

Vector symbolic architectures (VSAs) are high-dimensional vector representations of objects (e.g., words, image parts), relations (e.g., sentence structures), and sequences for use with machine learning algorithms. They consist of a vector addition operator for representing a collection of unordered objects, a binding operator for associating groups of objects, and a methodology for encoding complex structures. We first develop constraints that machine learning imposes on VSAs; for example, similar structures must be represented by similar vectors. The constraints suggest that current VSAs should represent phrases ("The smart Brazilian girl") by binding sums of terms, in addition to simply binding the terms directly. We show that matrix multiplication can be used as the binding operator for a VSA, and that matrix elements can be chosen at random. A consequence for living systems is that binding is mathematically possible without the need to specify, in advance, precise neuron-to-neuron connection properties for large numbers of synapses. A VSA that incorporates these ideas, Matrix Binding of Additive Terms (MBAT), is described that satisfies all

constraints. With respect to machine learning, for some types of problems appropriate VSA representations permit us to prove learnability rather than relying on simulations. We also propose dividing machine (and neural) learning and representation into three stages, with differing roles for learning in each stage. For neural modeling, we give representational reasons for nervous systems to have many recurrent connections, as well as for the importance of phrases in language processing. Sizing simulations and analyses suggest that VSAs in general, and MBAT in particular, are ready for real-world applications.

---

63. The Evolution of Representation in Simple Cognitive Networks

Lars Marstaller, Arend Hintze, Christoph Adami

https://doi.org/10.1162/neco_a_00475 | pp. 2079-2107

Representations are internal models of the environment that can provide guidance to a behaving agent, even in the absence of sensory information. It is not clear how representations are developed and whether they are necessary or even essential for intelligent behavior. We argue here that the ability to represent relevant features of the environment is the expected consequence of an adaptive process, give a formal definition of representation based on information theory, and quantify it with a measure R. To measure how R changes over time, we evolve two types of networks--an artificial neural network and a network of hidden Markov gates--to solve a categorization task using a genetic algorithm. We find that the capacity to represent increases during evolutionary adaptation and that agents form representations of their environment during their lifetime. This ability allows the agents to act on sensorial inputs in the context of their acquired representations and enables complex and context-dependent behavior. We examine which concepts (features of the environment) our networks are representing, how the representations are logically encoded in the networks, and how they form as an agent behaves to solve a task. We conclude that R should be able to quantify the representations within any cognitive system and should be predictive of an agent's long-term adaptive success.

---

64. Dynamics of Multiple-Choice Decision Making

Hongzhi You, Da■Hui Wang

https://doi.org/10.1162/neco_a_00473 | pp. 2108-2145

Neuroscientists have carried out comprehensive experiments to reveal the neural mechanisms underlying the perceptual decision making that pervades daily life. These experiments have illuminated salient features of decision making, including probabilistic choice behavior, the ramping activity of decision-related neurons, and the dependence of decision time and accuracy on the difficulty of the task. Spiking network models have reproduced these features, and a two-dimensional mean field model has demonstrated that the saddle node structure underlies two-alternative decision making. Here, we reduced a spiking network model to an analytically tractable, partial integro-differential system and characterized not only multiple-choice decision behaviors but also the time course of neural activities underlying decisions, providing a mechanistic explanation for the observations noted in the experiments. First, we observed that a two-bump unstable steady state of the system is responsible for two-choice decision making, similar to the saddle node structure in the two-dimensional mean field model. However, for four-choice decision making, three types of unstable steady states collectively predominate the time course of the evolution from the initial state to the stable states. Second, the time constant of the unstable steady state can explain the fact that four-choice

decision making requires a longer time than two-choice decision making. However, the quicker decision, given a stronger motion strength, cannot be explained by the time constant of the unstable steady state. Rather, the decision time can be attributed to the projection coefficient of the difference between the initial state and the unstable steady state on the eigenvector corresponding to the largest positive eigenvalue.

---

65. EEG Data Space Adaptation to Reduce Intersession Nonstationarity in Brain-Computer Interface

Mahnaz Arvaneh, Cuntai Guan, Kai Keng Ang, Chai Quek

https://doi.org/10.1162/neco_a_00474 | pp. 2146-2171

A major challenge in EEG-based brain-computer interfaces (BCIs) is the intersession nonstationarity in the EEG data that often leads to deteriorated BCI performances. To address this issue, this letter proposes a novel data space adaptation technique, EEG data space adaptation (EEG-DSA), to linearly transform the EEG data from the target space (evaluation session), such that the distribution difference to the source space (training session) is minimized. Using the Kullback-Leibler (KL) divergence criterion, we propose two versions of the EEG-DSA algorithm: the supervised version, when labeled data are available in the evaluation session, and the unsupervised version, when labeled data are not available. The performance of the proposed EEG-DSA algorithm is evaluated on the publicly available BCI Competition IV data set IIa and a data set recorded from 16 subjects performing motor imagery tasks on different days. The results show that the proposed EEG-DSA algorithm in both the supervised and unsupervised versions significantly outperforms the results without adaptation in terms of classification accuracy. The results also show that for subjects with poor BCI performances when no adaptation is applied, the proposed EEG-DSA algorithm in both the supervised and unsupervised versions significantly outperforms the unsupervised bias adaptation algorithm (PMean).

---

66. Alternating Direction Methods for Latent Variable Gaussian Graphical Model Selection

Shiqian Ma, Lingzhou Xue, Hui Zou

https://doi.org/10.1162/neco_a_00379 | pp. 2172-2198

Chandrasekaran, Parrilo, and Willsky (2012) proposed a convex optimization problem for graphical model selection in the presence of unobserved variables. This convex optimization problem aims to estimate an inverse covariance matrix that can be decomposed into a sparse matrix minus a low-rank matrix from sample data. Solving this convex optimization problem is very challenging, especially for large problems. In this letter, we propose two alternating direction methods for solving this problem. The first method is to apply the classic alternating direction method of multipliers to solve the problem as a consensus problem. The second method is a proximal gradient-based alternating-direction method of multipliers. Our methods take advantage of the special structure of the problem and thus can solve large problems very efficiently. A global convergence result is established for the proposed methods. Numerical results on both synthetic data and gene expression data show that our methods usually solve problems with 1 million variables in 1 to 2 minutes and are usually 5 to 35 times faster than a state-of-the-art Newton-CG proximal point algorithm.

---

67. A Monte Carlo Metropolis-Hastings Algorithm for Sampling from Distributions with Intractable Normalizing Constants

Faming Liang, Ick Hoon Jin

https://doi.org/10.1162/neco_a_00466 | pp. 2199-2234

Simulating from distributions with intractable normalizing constants has been a long-standing problem in machine learning. In this letter, we propose a new algorithm, the Monte Carlo Metropolis-Hastings (MCMH) algorithm, for tackling this problem. The MCMH algorithm is a Monte Carlo version of the Metropolis-Hastings algorithm. It replaces the unknown normalizing constant ratio by a Monte Carlo estimate in simulations, while still converges, as shown in the letter, to the desired target distribution under mild conditions. The MCMH algorithm is illustrated with spatial autologistic models and exponential random graph models. Unlike other auxiliary variable Markov chain Monte Carlo (MCMC) algorithms, such as the Møller and exchange algorithms, the MCMH algorithm avoids the requirement for perfect sampling, and thus can be applied to many statistical models for which perfect sampling is not available or very expensive. The MCMH algorithm can also be applied to Bayesian inference for random effect models and missing data problems that involve simulations from a distribution with intractable integrals.

---

68. Population Coding and the Labeling Problem: Extrinsic Versus Intrinsic Representations

Sidney R. Lehky, Margaret E. Sereno, Anne B. Sereno

https://doi.org/10.1162/neco_a_00486 | pp. 2235-2264

Current population coding methods, including weighted averaging and Bayesian estimation, are based on extrinsic representations. These require that neurons be labeled with response parameters, such as tuning curve peaks or noise distributions, which are tied to some external, world-based metric scale. Firing rates alone, without this external labeling, are insufficient to represent a variable. However, the extrinsic approach does not explain how such neural labeling is implemented. A radically different and perhaps more physiological approach is based on intrinsic representations, which have access only to firing rates. Because neurons are unlabeled, intrinsic coding represents relative, rather than absolute, values of a variable. We show that intrinsic coding has representational advantages, including invariance, categorization, and discrimination, and in certain situations it may also recover absolute stimulus values.

---

69. Information Transfer Through Stochastic Transmission of a Linear Combination of Rates

Ioannis Smyrnakis, Stelios M. Smirnakis

https://doi.org/10.1162/neco_a_00476 | pp. 2265-2302

In this work, the Shannon information transfer rate due to the transmission of a linear combination of the firing rates of a number of afferent neurons is examined. The transmission of this linear combination (transfer statistic) takes place through a stochastic firing process, while a rate code is assumed. Constraints are imposed on the transmission process by the requirement that the coefficient of variation for the transfer statistic is small and by the relative variance of the individual terms in the calculation of the statistic. In the regime of no noise or signal correlations among the input neurons, simulations suggest that information

transfer for fixed overall input is favored when there are few high-firing neurons, as opposed to more lower-firing neurons. Signal correlations among low-firing neurons can result in aggregates of high firing rates, improving in this way information transfer and calculational robustness. Under reasonable rate code assumptions, information transfer rates obtained are of the order 3 to 10 bit/sec.

---

70. Spike-Based Probabilistic Inference in Analog Graphical Models Using Interspike-Interval Coding

Andreas Steimer, Rodney J. Douglas

https://doi.org/10.1162/neco_a_00477 | pp. 2303-2354

Temporal spike codes play a crucial role in neural information processing. In particular, there is strong experimental evidence that interspike intervals (ISIs) are used for stimulus representation in neural systems. However, very few algorithmic principles exploit the benefits of such temporal codes for probabilistic inference of stimuli or decisions. Here, we describe and rigorously prove the functional properties of a spike-based processor that uses ISI distributions to perform probabilistic inference. The abstract processor architecture serves as a building block for more concrete, neural implementations of the belief-propagation (BP) algorithm in arbitrary graphical models (e.g., Bayesian networks and factor graphs). The distributed nature of graphical models matches well with the architectural and functional constraints imposed by biology. In our model, ISI distributions represent the BP messages exchanged between factor nodes, leading to the interpretation of a single spike as a random sample that follows such a distribution. We verify the abstract processor model by numerical simulation in full graphs, and demonstrate that it can be applied even in the presence of analog variables. As a particular example, we also show results of a concrete, neural implementation of the processor, although in principle our approach is more flexible and allows different neurobiological interpretations. Furthermore, electrophysiological data from area LIP during behavioral experiments are assessed in light of ISI coding, leading to concrete testable, quantitative predictions and a more accurate description of these data compared to hitherto existing models.

---

71. An Investigation of the Stochastic Hodgkin-Huxley Models Under Noisy Rate Functions

Marifi Güler

https://doi.org/10.1162/neco_a_00487 | pp. 2355-2372

The effects of ion channel fluctuations on the transmembrane voltage activity are potentially profound in small-size excitable membrane patches. Different groups have extended Hodgkin-Huxley equations into stochastic differential equations to capture the effects of ion channel noise analytically (Fox & Lu, 1994; Linaro, Storace, & Giugliano, 2011; Güler, 2013). Studies have shown that the accuracy of spiking statistics by Fox and Lu's model does not match well with the corresponding statistics from the exact microscopic simulations. The models of both Linaro et al. and Güler, however, were found to produce highly accurate statistics. Here we extend the examination of these models to the case in which the rate functions for the opening and closing of gates are under the influence of noise. For that purpose, the usual rate functions are accompanied additively by Ornstein-Uhlenbeck-type stochastic angular variables. Moreover, we argue that the existence of such noise in the rate functions is a plausible physiological phenomenon for finite-size membranes. It is observed that the presence of noise in the rates is not effective on the degree of

inaccuracies within the Fox and Lu model. Güler model's accuracy is found to remain high as in the case of noise free rates. But the performance of Linaro et al.'s model is seen to degrade seriously with the increasing strength of the introduced rate function noise. We attribute this failure of Linaro et al.'s model to the use of the covariance function of open channels at the steady state, in its derivation.

---

72. Dynamic Analysis of Naive Adaptive Brain-Machine Interfaces

Kevin Kowalski, Bryan He, Lakshminarayan Srinivasan

https://doi.org/10.1162/neco_a_00484 | pp. 2373-2420

The closed-loop operation of brain-machine interfaces (BMI) provides a context to discover foundational principles behind human-computer interaction, with emerging clinical applications to stroke, neuromuscular diseases, and trauma. In the canonical BMI, a user controls a prosthetic limb through neural signals that are recorded by electrodes and processed by a decoder into limb movements. In laboratory demonstrations with able-bodied test subjects, parameters of the decoder are commonly tuned using training data that include neural signals and corresponding overt arm movements. In the application of BMI to paralysis or amputation, arm movements are not feasible, and imagined movements create weaker, partially unrelated patterns of neural activity. BMI training must begin naive, without access to these prototypical methods for parameter initialization used in most laboratory BMI demonstrations. Naive adaptive BMI refer to a class of methods recently introduced to address this problem. We first identify the basic elements of existing approaches based on adaptive filtering and define a decoder, ReFIT-PPF to represent these existing approaches. We then present Joint RSE, a novel approach that logically extends prior approaches. Using recently developed human- and synthetic-subjects closed-loop BMI simulation platforms, we show that Joint RSE significantly outperforms ReFIT-PPF and nonadaptive (static) decoders. Control experiments demonstrate the critical role of jointly estimating neural parameters and user intent. In addition, we show that nonzero sensorimotor delay in the user significantly degrades ReFIT-PPF but not Joint RSE, owing to differences in the prior on intended velocity. Paradoxically, substantial differences in the nature of sensory feedback between these methods do not contribute to differences in performance between Joint RSE and ReFIT-PPF. Instead, BMI performance improvement is driven by machine learning, which outpaces rates of human learning in the human-subjects simulation platform. In this regime, nuances of error-related feedback to the human user are less relevant to rapid BMI mastery.

---

73. A Bio-Inspired, Computational Model Suggests Velocity Gradients of Optic Flow Locally Encode Ordinal Depth at Surface Borders and Globally They Encode Self-Motion

Florian Raudies, Stefan Ringbauer, Heiko Neumann

https://doi.org/10.1162/neco_a_00479 | pp. 2421-2449

Visual navigation requires the estimation of self-motion as well as the segmentation of objects from the background. We suggest a definition of local velocity gradients to compute types of self-motion, segment objects, and compute local properties of optical flow fields, such as divergence, curl, and shear. Such velocity gradients are computed as velocity differences measured locally tangent and normal to the direction of flow. Then these differences are rotated according to the local direction of flow to achieve independence of that direction. We propose a bio-inspired model for the computation of these velocity gradients for video

sequences. Simulation results show that local gradients encode ordinal surface depth, assuming self-motion in a rigid scene or object motions in a nonrigid scene. For translational self-motion velocity, gradients can be used to distinguish between static and moving objects. The information about ordinal surface depth and self-motion can help steering control for visual navigation.

---

74. Exploitation of Pairwise Class Distances for Ordinal Classification

Javier Sánchez■Monedero, Pedro Antonio Gutiérrez, Peter Ti■o, César Hervás■Martínez

https://doi.org/10.1162/neco_a_00478 | pp. 2450-2485

Ordinal classification refers to classification problems in which the classes have a natural order imposed on them because of the nature of the concept studied. Some ordinal classification approaches perform a projection from the input space to one-dimensional (latent) space that is partitioned into a sequence of intervals (one for each class). Class identity of a novel input pattern is then decided based on the interval its projection falls into. This projection is trained only indirectly as part of the overall model fitting. As with any other latent model fitting, direct construction hints one may have about the desired form of the latent model can prove very useful for obtaining high-quality models. The key idea of this letter is to construct such a projection model directly, using insights about the class distribution obtained from pairwise distance calculations. The proposed approach is extensively evaluated with 8 nominal and ordinal classifiers methods, 10 real-world ordinal classification data sets, and 4 different performance measures. The new methodology obtained the best results in average ranking when considering three of the performance metrics, although significant differences are found for only some of the methods. Also, after observing other methods of internal behavior in the latent space, we conclude that the internal projections do not fully reflect the intraclass behavior of the patterns. Our method is intrinsically simple, intuitive, and easily understandable, yet highly competitive with state-of-the-art approaches to ordinal classification.

---

75. Spherical Mesh Adaptive Direct Search for Separating Quasi-Uncorrelated Sources by Range-Based Independent Component Analysis

S. Easter Selvan, P. Borckmans, Amit Chattopadhyay, Pierre-Antoine Absil

https://doi.org/10.1162/neco_a_00485 | pp. 2486-2522

It is seemingly paradoxical to the classical definition of the independent component analysis (ICA), that in reality, the true sources are often not strictly uncorrelated. With this in mind, this letter concerns a framework to extract quasi-uncorrelated sources with finite supports by optimizing a range-based contrast function under unit-norm constraints (to handle the inherent scaling indeterminacy of ICA) but without orthogonality constraints. Albeit the appealing contrast properties of the range-based function (e.g., the absence of mixing local optima), the function is not differentiable everywhere. Unfortunately, there is a dearth of literature on derivative-free optimizers that effectively handle such a nonsmooth yet promising contrast function. This is the compelling reason for the design of a nonsmooth optimization algorithm on a manifold of matrices having unit-norm columns with the following objectives: to ascertain convergence to a Clarke stationary point of the contrast function and adhere to the necessary unit-norm constraints more naturally. The proposed nonsmooth optimization algorithm crucially relies on the design and analysis of an extension of the mesh adaptive direct search (MADS) method to handle locally Lipschitz objective functions defined on the sphere. The applicability of the algorithm in the ICA domain is demonstrated with simulations involving natural, face,

aerial, and texture images.

---

76. Scaling Laws of Associative Memory Retrieval

Sandro Romani, Itai Pinkoviezky, Alon Rubin, Misha Tsodyks

https://doi.org/10.1162/neco_a_00499 | pp. 2523-2544

Most people have great difficulty in recalling unrelated items. For example, in free recall experiments, lists of more than a few randomly selected words cannot be accurately repeated. Here we introduce a phenomenological model of memory retrieval inspired by theories of neuronal population coding of information. The model predicts nontrivial scaling behaviors for the mean and standard deviation of the number of recalled words for lists of increasing length. Our results suggest that associative information retrieval is a dominating factor that limits the number of recalled items.

---

77. Effect of Phase Response Curve Skewness on Synchronization of Electrically Coupled Neuronal Oscillators

Ramana Dodla, Charles J. Wilson

https://doi.org/10.1162/neco_a_00488 | pp. 2545-2610

We investigate why electrically coupled neuronal oscillators synchronize or fail to synchronize using the theory of weakly coupled oscillators. Stability of synchrony and antisynchrony is predicted analytically and verified using numerical bifurcation diagrams. The shape of the phase response curve (PRC), the shape of the voltage time course, and the frequency of spiking are freely varied to map out regions of parameter spaces that hold stable solutions. We find that type 1 and type 2 PRCs can hold both synchronous and antisynchronous solutions, but the shape of the PRC and the voltage determine the extent of their stability. This is achieved by introducing a five-piecewise linear model to the PRC and a three-piecewise linear model to the voltage time course, and then analyzing the resultant eigenvalue equations that determine the stability of the phase-locked solutions. A single time parameter defines the skewness of the PRC, and another single time parameter defines the spike width and frequency. Our approach gives a comprehensive picture of the relation of the PRC shape, voltage time course, and stability of the resultant synchronous and antisynchronous solutions.

---

78. Spike-Timing-Dependent Construction

Toby Lightheart, Steven Grainger, Tien■Fu Lu

https://doi.org/10.1162/neco_a_00501 | pp. 2611-2645

Spike-timing-dependent construction (STDC) is the production of new spiking neurons and connections in a simulated neural network in response to neuron activity. Following the discovery of spike-timing-dependent plasticity (STDP), significant effort has gone into the modeling and simulation of adaptation in spiking neural networks (SNNs). Limitations in computational power imposed by network topology, however, constrain learning capabilities through connection weight modification alone. Constructive algorithms produce new neurons and connections, allowing automatic structural responses for applications of unknown complexity and nonstationary solutions. A conceptual analogy is developed and extended to theoretical conditions for

modeling synaptic plasticity as network construction. Generalizing past constructive algorithms, we propose a framework for the design of novel constructive SNNs and demonstrate its application in the development of simulations for the validation of developed theory. Potential directions of future research and applications of STDC for biological modeling and machine learning are also discussed.

---

79. Computer Modeling of Mild Axonal Injury: Implications for Axonal Signal Transmission

Vladislav Volman, Laurel J. Ng

https://doi.org/10.1162/neco_a_00491 | pp. 2646-2681

Diffusion imaging and postmortem studies of patients with mild traumatic brain injury (mTBI) of the concussive type are consistent with the observations of diffuse axonal injury to the white matter axons. Mechanical trauma to axons affects the properties of tetrodotoxin-sensitive sodium channels at the nodes of Ranvier, leading to axonal degeneration through intra-axonal accumulation of calcium ions and activation of calcium proteases; however, the immediate implications of axonal trauma regarding axonal functionality and their relevance to transient impairment of function as observed in concussion remain elusive. A biophysically realistic computational model of a myelinated axon was developed to investigate how mTBI could immediately affect axonal function. Traumatized axons showed alterations in signal propagation properties that nonlinearly depended on the level of trauma; subthreshold traumatized axons had decreased spike propagation time, whereas suprathreshold traumatized axons exhibited a slowdown of spike propagation and spike propagation failure. Trauma had consistently reduced axonal spike amplitude. The susceptibility of an axon to trauma could be modulated by the function of an ATP-dependent sodium-potassium pump. The results suggest a mechanism by which concussive mTBI could lead to the immediate impairment of signal propagation through the axon and the emerging dysfunctional neuronal information exchange.

---

80. A Principled Dimension-Reduction Method for the Population Density Approach to Modeling Networks of Neurons with Synaptic Dynamics

Cheng Ly

https://doi.org/10.1162/neco_a_00489 | pp. 2682-2708

The population density approach to neural network modeling has been utilized in a variety of contexts. The idea is to group many similar noisy neurons into populations and track the probability density function for each population that encompasses the proportion of neurons with a particular state rather than simulating individual neurons (i.e., Monte Carlo). It is commonly used for both analytic insight and as a time-saving computational tool. The main shortcoming of this method is that when realistic attributes are incorporated in the underlying neuron model, the dimension of the probability density function increases, leading to intractable equations or, at best, computationally intensive simulations. Thus, developing principled dimension-reduction methods is essential for the robustness of these powerful methods. As a more pragmatic tool, it would be of great value for the larger theoretical neuroscience community. For exposition of this method, we consider a single uncoupled population of leaky integrate-and-fire neurons receiving external excitatory synaptic input only. We present a dimension-reduction method that reduces a two-dimensional partial differential-integral equation to a computationally efficient one-dimensional system and gives qualitatively accurate results in both the steady-state and nonequilibrium regimes. The method,

termed modified mean-field method, is based entirely on the governing equations and not on any auxiliary variables or parameters, and it does not require fine-tuning. The principles of the modified mean-field method have potential applicability to more realistic (i.e., higher-dimensional) neural networks.

---

81. Discriminative Learning of Propagation and Spatial Pattern for Motor Imagery EEG Analysis

Xinyang Li, Haihong Zhang, Cuntai Guan, Sim Heng Ong, Kai Keng Ang, Yaozhang Pan

https://doi.org/10.1162/neco_a_00500 | pp. 2709-2733

Effective learning and recovery of relevant source brain activity patterns is a major challenge to brain-computer interface using scalp EEG. Various spatial filtering solutions have been developed. Most current methods estimate an instantaneous demixing with the assumption of uncorrelatedness of the source signals. However, recent evidence in neuroscience suggests that multiple brain regions cooperate, especially during motor imagery, a major modality of brain activity for brain-computer interface. In this sense, methods that assume uncorrelatedness of the sources become inaccurate. Therefore, we are promoting a new methodology that considers both volume conduction effect and signal propagation between multiple brain regions. Specifically, we propose a novel discriminative algorithm for joint learning of propagation and spatial pattern with an iterative optimization solution. To validate the new methodology, we conduct experiments involving 16 healthy subjects and perform numerical analysis of the proposed algorithm for EEG classification in motor imagery brain-computer interface. Results from extensive analysis validate the effectiveness of the new methodology with high statistical significance.

---

82. Density-Difference Estimation

Masashi Sugiyama, Takafumi Kanamori, Taiji Suzuki, Marthinus Christoffel du Plessis, Song Liu, Ichiro Takeuchi

https://doi.org/10.1162/neco_a_00492 | pp. 2734-2775

We address the problem of estimating the difference between two probability densities. A naive approach is a two-step procedure of first estimating two densities separately and then computing their difference. However, this procedure does not necessarily work well because the first step is performed without regard to the second step, and thus a small estimation error incurred in the first stage can cause a big error in the second stage. In this letter, we propose a single-shot procedure for directly estimating the density difference without separately estimating two densities. We derive a nonparametric finite-sample error bound for the proposed single-shot density-difference estimator and show that it achieves the optimal convergence rate. We then show how the proposed density-difference estimator can be used in L^2-distance approximation. Finally, we experimentally demonstrate the usefulness of the proposed method in robust distribution comparison such as class-prior estimation and change-point detection.

---

83. Block Clustering Based on Difference of Convex Functions (DC) Programming and DC Algorithms

Hoai Minh Le, Hoai An Le Thi, Tao Pham Dinh, Van Ngai Huynh

https://doi.org/10.1162/neco_a_00490 | pp. 2776-2807

We investigate difference of convex functions (DC) programming and the DC algorithm (DCA) to solve the block clustering problem in the continuous framework, which traditionally requires solving a hard combinatorial optimization problem. DC reformulation techniques and exact penalty in DC programming are developed to build an appropriate equivalent DC program of the block clustering problem. They lead to an elegant and explicit DCA scheme for the resulting DC program. Computational experiments show the robustness and efficiency of the proposed algorithm and its superiority over standard algorithms such as two-mode K-means, two-mode fuzzy clustering, and block classification EM.

---

84. What Is the Limit of Redundancy Reduction with Divisive Normalization?

Fabian H. Sinz, Matthias Bethge

https://doi.org/10.1162/neco_a_00505 | pp. 2809-2814

Divisive normalization has been proposed as a nonlinear redundancy reduction mechanism capturing contrast correlations. Its basic function is a radial rescaling of the population response. Because of the saturation of divisive normalization, however, it is impossible to achieve a fully independent representation. In this letter, we derive an analytical upper bound on the inevitable residual redundancy of any saturating radial rescaling mechanism.

---

85. A Biological Gradient Descent for Prediction Through a Combination of STDP and Homeostatic Plasticity

Mathieu Galtier, Gilles Wainrib

https://doi.org/10.1162/neco_a_00512 | pp. 2815-2832

Identifying, formalizing, and combining biological mechanisms that implement known brain functions, such as prediction, is a main aspect of research in theoretical neuroscience. In this letter, the mechanisms of spike-timing-dependent plasticity and homeostatic plasticity, combined in an original mathematical formalism, are shown to shape recurrent neural networks into predictors. Following a rigorous mathematical treatment, we prove that they implement the online gradient descent of a distance between the network activity and its stimuli. The convergence to an equilibrium, where the network can spontaneously reproduce or predict its stimuli, does not suffer from bifurcation issues usually encountered in learning in recurrent neural networks.

---

86. Formation and Regulation of Dynamic Patterns in Two-Dimensional Spiking Neural Circuits with Spike-Timing-Dependent Plasticity

John H. Palmer, Pulin Gong

https://doi.org/10.1162/neco_a_00511 | pp. 2833-2857

Spike-timing-dependent plasticity (STDP) is an important synaptic dynamics that is capable of shaping the complex spatiotemporal activity of neural circuits. In this study, we examine the effects of STDP on the spatiotemporal patterns of a spatially extended, two-dimensional spiking neural circuit. We show that STDP

can promote the formation of multiple, localized spiking wave patterns or multiple spike timing sequences in a broad parameter space of the neural circuit. Furthermore, we illustrate that the formation of these dynamic patterns is due to the interaction between the dynamics of ongoing patterns in the neural circuit and STDP. This interaction is analyzed by developing a simple model able to capture its essential dynamics, which give rise to symmetry breaking. This occurs in a fundamentally self-organizing manner, without fine-tuning of the system parameters. Moreover, we find that STDP provides a synaptic mechanism to learn the paths taken by spiking waves and modulate the dynamics of their interactions, enabling them to be regulated. This regulation mechanism has error-correcting properties. Our results therefore highlight the important roles played by STDP in facilitating the formation and regulation of spiking wave patterns that may have crucial functional roles in brain information processing.

---

87. Encoding Binary Neural Codes in Networks of Threshold-Linear Neurons

Carina Curto, Anda Degeratu, Vladimir Itskov

https://doi.org/10.1162/neco_a_00504 | pp. 2858-2903

Networks of neurons in the brain encode preferred patterns of neural activity via their synaptic connections. Despite receiving considerable attention, the precise relationship between network connectivity and encoded patterns is still poorly understood. Here we consider this problem for networks of threshold-linear neurons whose computational function is to learn and store a set of binary patterns (e.g., a neural code) as "permitted sets" of the network. We introduce a simple encoding rule that selectively turns "on" synapses between neurons that coappear in one or more patterns. The rule uses synapses that are binary, in the sense of having only two states ("on" or "off"), but also heterogeneous, with weights drawn from an underlying synaptic strength matrix S. Our main results precisely describe the stored patterns that result from the encoding rule, including unintended "spurious" states, and give an explicit characterization of the dependence on S. In particular, we find that binary patterns are successfully stored in these networks when the excitatory connections between neurons are geometrically balanced--i.e., they satisfy a set of geometric constraints. Furthermore, we find that certain types of neural codes are natural in the context of these networks, meaning that the full code can be accurately learned from a highly undersampled set of patterns. Interestingly, many commonly observed neural codes in cortical and hippocampal areas are natural in this sense. As an application, we construct networks that encode hippocampal place field codes nearly exactly, following presentation of only a small fraction of patterns. To obtain our results, we prove new theorems using classical ideas from convex and distance geometry, such as Cayley-Menger determinants, revealing a novel connection between these areas of mathematics and coding properties of neural networks.

---

88. Attention as Reward-Driven Optimization of Sensory Processing

Matthew Chalk, Iain Murray, Peggy Seriès

https://doi.org/10.1162/neco_a_00494 | pp. 2904-2933

Attention causes diverse changes to visual neuron responses, including alterations in receptive field structure, and firing rates. A common theoretical approach to investigate why sensory neurons behave as they do is based on the efficient coding hypothesis: that sensory processing is optimized toward the statistics of the received input. We extend this approach to account for the influence of task demands, hypothesizing that the brain learns a probabilistic model of both the sensory input and reward received for

performing different actions. Attention-dependent changes to neural responses reflect optimization of this internal model to deal with changes in the sensory environment (stimulus statistics) and behavioral demands (reward statistics). We use this framework to construct a simple model of visual processing that is able to replicate a number of attention-dependent changes to the responses of neurons in the midlevel visual cortices. The model is consistent with and provides a normative explanation for recent divisive normalization models of attention (Reynolds & Heeger, 2009).

---

89. Neuronal Assembly Dynamics in Supervised and Unsupervised Learning Scenarios

Renan C. Moioli, Phil Husbands

https://doi.org/10.1162/neco_a_00502 | pp. 2934-2975

The dynamic formation of groups of neurons--neuronal assemblies--is believed to mediate cognitive phenomena at many levels, but their detailed operation and mechanisms of interaction are still to be uncovered. One hypothesis suggests that synchronized oscillations underpin their formation and functioning, with a focus on the temporal structure of neuronal signals. In this context, we investigate neuronal assembly dynamics in two complementary scenarios: the first, a supervised spike pattern classification task, in which noisy variations of a collection of spikes have to be correctly labeled; the second, an unsupervised, minimally cognitive evolutionary robotics tasks, in which an evolved agent has to cope with multiple, possibly conflicting, objectives. In both cases, the more traditional dynamical analysis of the system's variables is paired with information-theoretic techniques in order to get a broader picture of the ongoing interactions with and within the network. The neural network model is inspired by the Kuramoto model of coupled phase oscillators and allows one to fine-tune the network synchronization dynamics and assembly configuration. The experiments explore the computational power, redundancy, and generalization capability of neuronal circuits, demonstrating that performance depends nonlinearly on the number of assemblies and neurons in the network and showing that the framework can be exploited to generate minimally cognitive behaviors, with dynamic assembly formation accounting for varying degrees of stimuli modulation of the sensorimotor interactions.

---

90. Formal Modeling of Robot Behavior with Learning

Ryan Fraser Kirwan, Alice Miller, Bernd Porr, Paolo Di Prodi

https://doi.org/10.1162/neco_a_00493 | pp. 2976-3019

We present formal specification and verification of a robot moving in a complex network, using temporal sequence learning to avoid obstacles. Our aim is to demonstrate the benefit of using a formal approach to analyze such a system as a complementary approach to simulation. We first describe a classical closed-loop simulation of the system and compare this approach to one in which the system is analyzed using formal verification. We show that the formal verification has some advantages over classical simulation and finds deficiencies our classical simulation did not identify. Specifically we present a formal specification of the system, defined in the Promela modeling language and show how the associated model is verified using the Spin model checker. We then introduce an abstract model that is suitable for verifying the same properties for any environment with obstacles under a given set of assumptions. We outline how we can prove that our abstraction is sound: any property that holds for the abstracted model will hold in the

original (unabstracted) model.

91. Toward Nonlinear Local Reinforcement Learning Rules Through Neuroevolution

Vassilis Vassiliades, Chris Christodoulou

https://doi.org/10.1162/neco_a_00514 | pp. 3020-3043

We consider the problem of designing local reinforcement learning rules for artificial neural network (ANN) controllers. Motivated by the universal approximation properties of ANNs, we adopt an ANN representation for the learning rules, which are optimized using evolutionary algorithms. We evaluate the ANN rules in partially observable versions of four tasks: the mountain car, the acrobot, the cart pole balancing, and the nonstationary mountain car. For testing whether such evolved ANN-based learning rules perform satisfactorily, we compare their performance with the performance of SARSA(λ) with tile coding, when the latter is provided with either full or partial state information. The comparison shows that the evolved rules perform much better than SARSA(λ) with partial state information and are comparable to the one with full state information, while in the case of the nonstationary environment, the evolved rule is much more adaptive. It is therefore clear that the proposed approach can be particularly effective in both partially observable and nonstationary environments. Moreover, it could potentially be utilized toward creating more general rules that can be applied in multiple domains and transfer learning scenarios.

92. Online Learning of Single- and Multivalued Functions with an Infinite Mixture of Linear Experts

Bruno Damas, José Santos-Victor

https://doi.org/10.1162/neco_a_00510 | pp. 3044-3091

We present a supervised learning algorithm for estimation of generic input-output relations in a real-time, online fashion. The proposed method is based on a generalized expectation-maximization approach to fit an infinite mixture of linear experts (IMLE) to an online stream of data samples. This probabilistic model, while not fully Bayesian, can efficiently choose the number of experts that are allocated to the mixture, this way effectively controlling the complexity of the resulting model. The result is an incremental, online, and localized learning algorithm that performs nonlinear, multivariate regression on multivariate outputs by approximating the target function by a linear relation within each expert input domain and that can allocate new experts as needed. A distinctive feature of the proposed method is the ability to learn multivalued functions: one-to-many mappings that naturally arise in some robotic and computer vision learning domains, using an approach based on a Bayesian generative model for the predictions provided by each of the mixture experts. As a consequence, it is able to directly provide forward and inverse relations from the same learned mixture model. We conduct an extensive set of experiments to evaluate the proposed algorithm performance, and the results show that it can outperform state-of-the-art online function approximation algorithms in single-valued regression, while demonstrating good estimation capabilities in a multivalued function approximation context.

93. Dynamical Regimes in Neural Network Models of Matching Behavior

Kiyohito Iigaya, Stefano Fusi

https://doi.org/10.1162/neco_a_00522 | pp. 3093-3112

The matching law constitutes a quantitative description of choice behavior that is often observed in foraging tasks. According to the matching law, organisms distribute their behavior across available response alternatives in the same proportion that reinforcers are distributed across those alternatives. Recently a few biophysically plausible neural network models have been proposed to explain the matching behavior observed in the experiments. Here we study systematically the learning dynamics of these networks while performing a matching task on the concurrent variable interval (VI) schedule. We found that the model neural network can operate in one of three qualitatively different regimes depending on the parameters that characterize the synaptic dynamics and the reward schedule: (1) a matching behavior regime, in which the probability of choosing an option is roughly proportional to the baiting fractional probability of that option; (2) a perseverative regime, in which the network tends to make always the same decision; and (3) a tristable regime, in which the network can either perseverate or choose the two targets randomly approximately with the same probability. Different parameters of the synaptic dynamics lead to different types of deviations from the matching law, some of which have been observed experimentally. We show that the performance of the network depends on the number of stable states of each synapse and that bistable synapses perform close to optimal when the proper learning rate is chosen. Because our model provides a link between synaptic dynamics and qualitatively different behaviors, this work provides us with insight into the effects of neuromodulators on adaptive behaviors and psychiatric disorders.

---

94. Supervised Spike-Timing-Dependent Plasticity: A Spatiotemporal Neuronal Learning Rule for Function Approximation and Decisions

Jan-Moritz P. Franosch, Sebastian Urban, J. Leo van Hemmen

https://doi.org/10.1162/neco_a_00520 | pp. 3113-3130

How can an animal learn from experience? How can it train sensors, such as the auditory or tactile system, based on other sensory input such as the visual system? Supervised spike-timing-dependent plasticity (supervised STDP) is a possible answer. Supervised STDP trains one modality using input from another one as "supervisor." Quite complex time-dependent relationships between the senses can be learned. Here we prove that under very general conditions, supervised STDP converges to a stable configuration of synaptic weights leading to a reconstruction of primary sensory input.

---

95. Spontaneous Slow Oscillations and Sequential Patterns Due to Short-Term Plasticity in a Model of the Cortex

Timothée Leleu, Kazuyuki Aihara

https://doi.org/10.1162/neco_a_00513 | pp. 3131-3182

We study a realistic model of a cortical column taking into account short-term plasticity between pyramidal cells and interneurons. The simulation of leaky integrate-and-fire neurons shows that low-frequency oscillations emerge spontaneously as a result of intrinsic network properties. These oscillations are composed of prolonged phases of high and low activity reminiscent of cortical up and down states,

respectively. We simplify the description of the network activity by using a mean field approximation and reduce the system to two slow variables exhibiting some relaxation oscillations. We identify two types of slow oscillations. When the combination of dynamic synapses between pyramidal cells and those between interneurons accounts for the generation of these slow oscillations, the end of the up phase is characterized by asynchronous fluctuations of the membrane potentials. When the slow oscillations are mainly driven by the dynamic synapses between interneurons, the network exhibits fluctuations of membrane potentials, which are more synchronous at the end than at the beginning of the up phase. Additionally, finite size effect and slow synaptic currents can modify the irregularity and frequency, respectively, of these oscillations. Finally, we consider possible roles of a slow oscillatory input modeling long-range interactions in the brain. Spontaneous slow oscillations of local networks are modulated by the oscillatory input, which induces, notably, synchronization, subharmonic synchronization, and chaotic relaxation oscillations in the mean field approximation. In the case of forced oscillations, the slow population-averaged activity of leaky integrate-and-fire neurons can have both deterministic and stochastic temporal features. We discuss the possibility that long-range connectivity controls the emergence of slow sequential patterns in local populations due to the tendency of a cortical column to oscillate at low frequency.

---

96. On the Continuous Differentiability of Inter-Spike Intervals of Synaptically Connected Cortical Spiking Neurons in a Neuronal Network

Gautam Kumar, Mayuresh V. Kothare

https://doi.org/10.1162/neco_a_00503 | pp. 3183-3206

We derive conditions for continuous differentiability of inter-spike intervals (ISIs) of spiking neurons with respect to parameters (decision variables) of an external stimulating input current that drives a recurrent network of synaptically connected neurons. The dynamical behavior of individual neurons is represented by a class of discontinuous single-neuron models. We report here that ISIs of neurons in the network are continuously differentiable with respect to decision variables if (1) a continuously differentiable trajectory of the membrane potential exists between consecutive action potentials with respect to time and decision variables and (2) the partial derivative of the membrane potential of spiking neurons with respect to time is not equal to the partial derivative of their firing threshold with respect to time at the time of action potentials. Our theoretical results are supported by showing fulfillment of these conditions for a class of known bidimensional spiking neuron models.

---

97. Complete Classification of the Macroscopic Behavior of a Heterogeneous Network of Theta Neurons

Tanushree B. Luke, Ernest Barreto, Paul So

https://doi.org/10.1162/neco_a_00525 | pp. 3207-3234

We design and analyze the dynamics of a large network of theta neurons, which are idealized type I neurons. The network is heterogeneous in that it includes both inherently spiking and excitable neurons. The coupling is global, via pulselike synapses of adjustable sharpness. Using recently developed analytical methods, we identify all possible asymptotic states that can be exhibited by a mean field variable that captures the network's macroscopic state. These consist of two equilibrium states that reflect partial synchronization in the network and a limit cycle state in which the degree of network synchronization

oscillates in time. Our approach also permits a complete bifurcation analysis, which we carry out with respect to parameters that capture the degree of excitability of the neurons, the heterogeneity in the population, and the coupling strength (which can be excitatory or inhibitory). We find that the network typically tends toward the two macroscopic equilibrium states when the neuron's intrinsic dynamics and the network interactions reinforce one another. In contrast, the limit cycle state, bifurcations, and multistability tend to occur when there is competition among these network features. Finally, we show that our results are exhibited by finite network realizations of reasonable size.

---

98. Deficient GABAergic Gliotransmission May Cause Broader Sensory Tuning in Schizophrenia

Osamu Hoshino

https://doi.org/10.1162/neco_a_00519 | pp. 3235-3262

We examined how the depression of intracortical inhibition due to a reduction in ambient GABA concentration impairs perceptual information processing in schizophrenia. A neural network model with a gliotransmission-mediated ambient GABA regulatory mechanism was simulated. In the network, interneuron-to-glial-cell and principal-cell-to-glial-cell synaptic contacts were made. The former hyperpolarized glial cells and let their transporters import (remove) GABA from the extracellular space, thereby lowering ambient GABA concentration, reducing extrasynaptic GABAa receptor-mediated tonic inhibitory current, and thus exciting principal cells. In contrast, the latter depolarized the glial cells and let the transporters export GABA into the extracellular space, thereby elevating the ambient GABA concentration and thus inhibiting the principal cells. A reduction in ambient GABA concentration was assumed for a schizophrenia network. Multiple dynamic cell assemblies were organized as sensory feature columns. Each cell assembly responded to one specific feature stimulus. The tuning performance of the network to an applied feature stimulus was evaluated in relation to the level of ambient GABA. Transporter-deficient glial cells caused a deficit in GABAergic gliotransmission and reduced ambient GABA concentration, which markedly deteriorated the tuning performance of the network, broadening the sensory tuning. Interestingly, the GABAergic gliotransmission mechanism could regulate local ambient GABA levels: it augmented ambient GABA around stimulus-irrelevant principal cells, while reducing ambient GABA around stimulus-relevant principal cells, thereby ensuring their selective responsiveness to the applied stimulus. We suggest that a deficit in GABAergic gliotransmission may cause a reduction in ambient GABA concentration, leading to a broadening of sensory tuning in schizophrenia. The GABAergic gliotransmission mechanism proposed here may have an important role in the regulation of local ambient GABA levels, thereby improving the sensory tuning performance of the cortex.

---

99. Reinforcement Learning of Two-Joint Virtual Arm Reaching in a Computer Model of Sensorimotor Cortex

Samuel A. Neymotin, George L. Chadderdon, Cliff C. Kerr, Joseph T. Francis, William W. Lytton

https://doi.org/10.1162/neco_a_00521 | pp. 3263-3293

Neocortical mechanisms of learning sensorimotor control involve a complex series of interactions at multiple levels, from synaptic mechanisms to cellular dynamics to network connectomics. We developed a model of sensory and motor neocortex consisting of 704 spiking model neurons. Sensory and motor populations

included excitatory cells and two types of interneurons. Neurons were interconnected with AMPA/NMDA and GABAA synapses. We trained our model using spike-timing-dependent reinforcement learning to control a two-joint virtual arm to reach to a fixed target. For each of 125 trained networks, we used 200 training sessions, each involving 15 s reaches to the target from 16 starting positions. Learning altered network dynamics, with enhancements to neuronal synchrony and behaviorally relevant information flow between neurons. After learning, networks demonstrated retention of behaviorally relevant memories by using proprioceptive information to perform reach-to-target from multiple starting positions. Networks dynamically controlled which joint rotations to use to reach a target, depending on current arm position. Learning-dependent network reorganization was evident in both sensory and motor populations: learned synaptic weights showed target-specific patterning optimized for particular reach movements. Our model embodies an integrative hypothesis of sensorimotor cortical learning that could be used to interpret future electrophysiological data recorded in vivo from sensorimotor learning experiments. We used our model to make the following predictions: learning enhances synchrony in neuronal populations and behaviorally relevant information flow across neuronal populations, enhanced sensory processing aids task-relevant motor performance and the relative ease of a particular movement in vivo depends on the amount of sensory information required to complete the movement.

---

100. Speech Emotional Features Extraction Based on Electroglottograph

Lijiang Chen, Xia Mao, Pengfei Wei, Angelo Compare

https://doi.org/10.1162/neco_a_00523 | pp. 3294-3317

This study proposes two classes of speech emotional features extracted from electroglottography (EGG) and speech signal. The power-law distribution coefficients (PLDC) of voiced segments duration, pitch rise duration, and pitch down duration are obtained to reflect the information of vocal folds excitation. The real discrete cosine transform coefficients of the normalized spectrum of EGG and speech signal are calculated to reflect the information of vocal tract modulation. Two experiments are carried out. One is of proposed features and traditional features based on sequential forward floating search and sequential backward floating search. The other is the comparative emotion recognition based on support vector machine. The results show that proposed features are better than those commonly used in the case of speaker-independent and content-independent speech emotion recognition.

---

101. Bayesian Sparse Partial Least Squares

Diego Vidaurre, Marcel van Gerven, Concha Bielza, Pedro Larrañaga, Tom Heskes

https://doi.org/10.1162/neco_a_00524 | pp. 3318-3339

Partial least squares (PLS) is a class of methods that makes use of a set of latent or unobserved variables to model the relation between (typically) two sets of input and output variables, respectively. Several flavors, depending on how the latent variables or components are computed, have been developed over the last years. In this letter, we propose a Bayesian formulation of PLS along with some extensions. In a nutshell, we provide sparsity at the input space level and an automatic estimation of the optimal number of latent components. We follow the variational approach to infer the parameter distributions. We have successfully tested the proposed methods on a synthetic data benchmark and on electrocorticogram data associated with several motor outputs in monkeys.

102. Some Remarks on “Almost Periodic Dynamics of a Class of Delayed Neural Networks with Discontinuous Activations”

Tianping Chen

https://doi.org/10.1162/neco_c_00526 | pp. 3340-3342

There are some incorrect uses of mathematical terminology and typos in previous papers. We rewrite them precisely here.In Lu, Wenlian, Chen, and Tianping (2008), we studied the following class of delayed differential equations: 1where di(t), aij(t), Ii(t), , are some functions; , , are some nondecreasing discontinuous functions for any , dKij(t, s); and , are Lebesgue-Stieljies measures with respect to s.We have noticed some misuse of mathematical terminology in this previous letter, and rewrite them precisely here.The following paragraph is from that letter (pp. 1075–1076): weakly converges to on space . is dense in the Banach space), which is the conjugate space). Therefore, for each , we have By the uniform essential boundedness of and the Lebesgue dominant convergence theorem, we conclude that weakly converges to on the space .It should be rewritten as: We claim weakly converges to on space .In fact, for each (noting that is dense in the Banach space , which is the dual space of L1([0, T], Rn)), we have By uniform boundedness of and Lebesgue dominant convergence theorem, we conclude that weakly converges to on the space .Similarly, on page 696 of Lu, Wenlian, Chen, and Tianping (2006), there is a problem in the following paragraph: For each (noticing that is dense in the Banach space L1([0, T], Rn)), holds and is uniformly bounded. Therefore, weakly converges to on .The correct wording is: For each (noticing that is dense in the Banach space , which is the dual space of L1([0, T], Rn)), holds and is the uniformly bounded. Therefore, weakly converges to on L1([0, T], Rn).Moreover, inequality 4.3 on page 1085 of Lu, Wenlian, Chen, and Tianping (2008), should be replaced by because it is a special case of inequality 3.1 on page 1072: